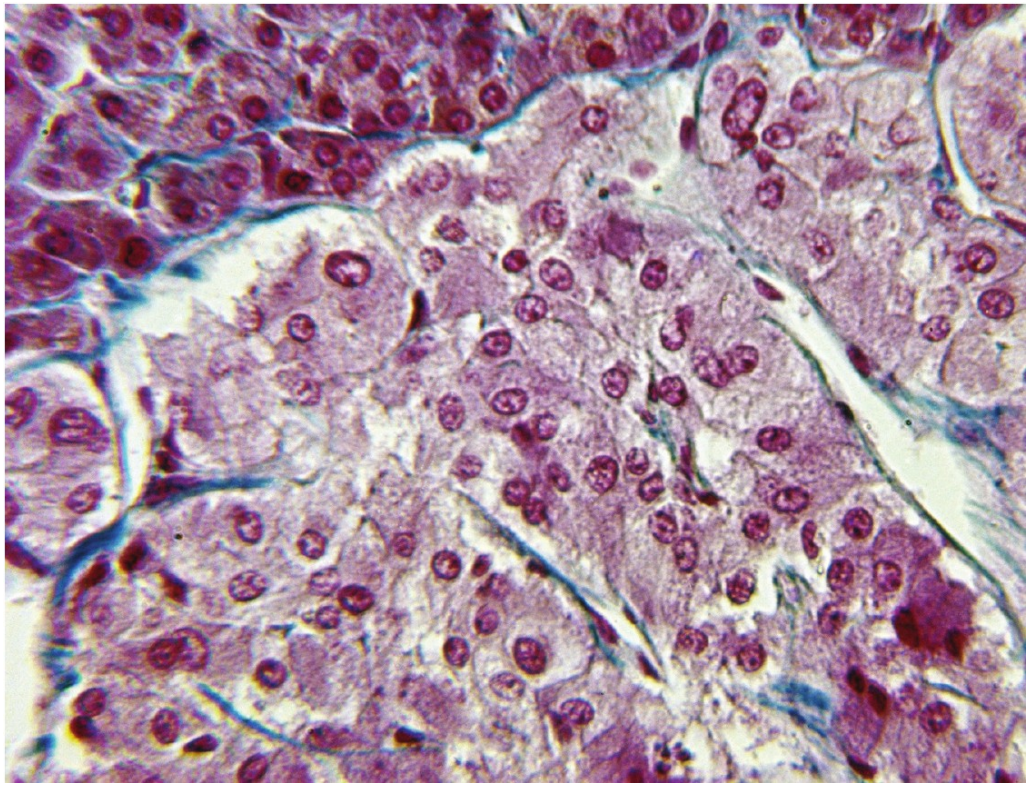


Carlson and Birkett (2022), Chapter 18 opener, p.580. Image: De Agostini Picture Library/Science Source.

BEHAVIORAL NEUROSCIENCE 392

Stress and Anxiety Disorders

When Threat Persists: Stress, Anxiety, and Recovery

Fall 2026

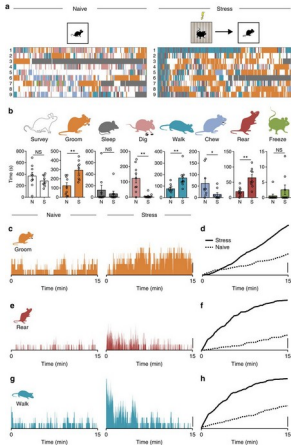
Jordan Lopez, Ph.D.
Professor of Neurobiology & Biomedical Engineering

Stress coordinates the body with a challenge

A stressor is a demand that threatens stability or requires adaptation. Stress responses change circulation, metabolism, attention, and behavior so that an animal can respond. Their usefulness depends on the challenge, how long it lasts, and whether the animal can return to ordinary activity.

The hypothalamus helps organize these responses through autonomic nerves and hormone release. Its paraventricular nucleus contains neurons that produce corticotropin-releasing hormone, or CRH. These cells participate in the hormonal stress response and also influence which actions follow a stressful event.

After acute stress, a control mouse alternates among exploration, grooming, and other behaviors. Grooming is a normal body-care action, not a diagnosis of anxiety. The order and duration of these actions provide observable measures of how the animal reorganizes behavior after the challenge ends.



Füzesi et al. (2016), Fig. 1.



Control mouse after acute stress. Füzesi et al. (2016).

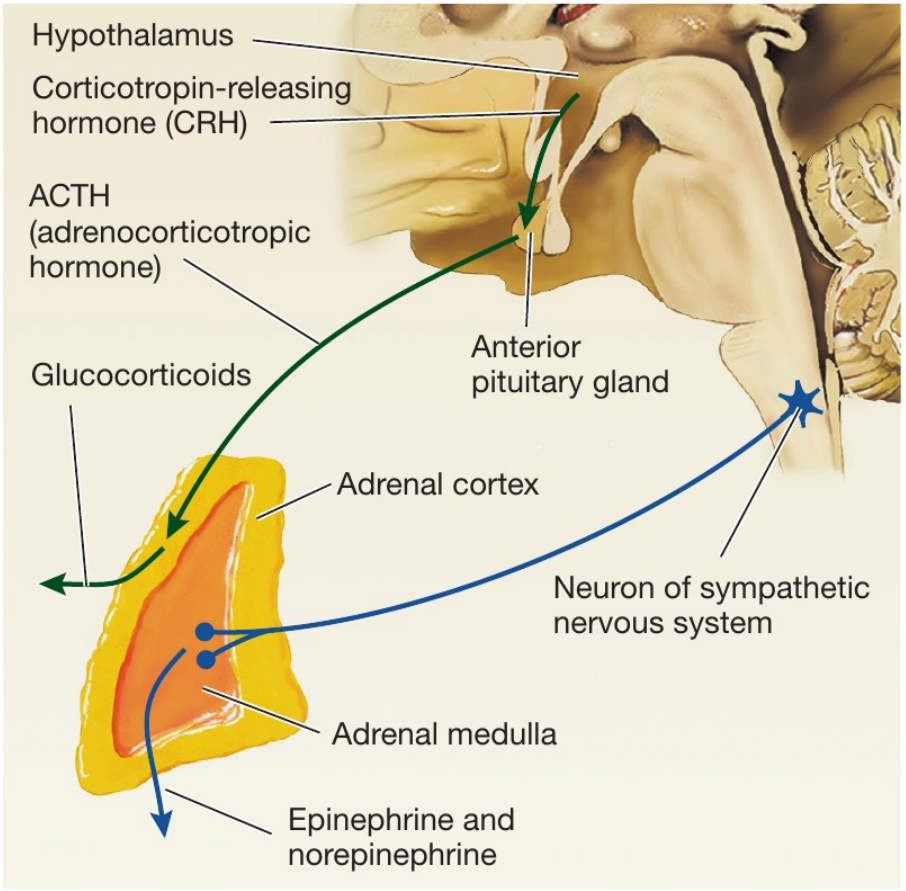
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Sympathetic nerves provide a rapid response

The sympathetic division of the autonomic nervous system rapidly changes organ activity. Its signals increase cardiac output and redirect resources toward immediate action. These nerve impulses can change physiology before a slower hormone response has developed throughout the circulation.

The adrenal medulla is the inner region of each adrenal gland. Sympathetic input causes its cells to release epinephrine and norepinephrine into blood. These catecholamines extend the response to tissues beyond the immediate nerve endings and help mobilize available energy.

Autonomic responses do not require one fixed behavior. An animal may freeze, flee, investigate, or groom while some physiological components overlap. Identifying the action separately from heart rate or hormone release avoids treating all responses to a stressor as the same measurement.



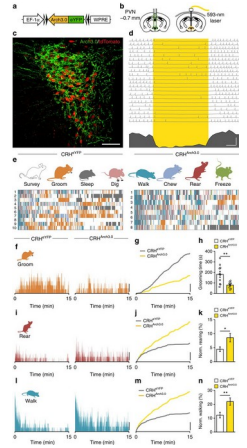
Carlson and Birkett (2022), Fig. 18.1, p.582.

Silencing CRH neurons changes post-stress behavior

Füzesi and colleagues manipulated CRH neurons in the hypothalamic paraventricular nucleus of mice. They used a light-sensitive inhibitory protein to reduce these cells' activity after acute stress. This allowed them to change the neurons during the period when behavior was being reorganized.

Silencing CRH neurons disrupted the normal pattern of post-stress actions. The investigators scored individual behaviors rather than assigning one general anxiety value to the session. A change in the arrangement of grooming and exploration therefore reflected a measurable alteration in the behavioral repertoire.

The original comparison follows mice with and without optical silencing. Light controls distinguish the effect of changing CRH-cell activity from the presence of illumination. This neuronal manipulation acts upstream of several possible outputs, so it is not equivalent to directly changing blood cortisol alone.



Füzesi et al. (2016), Fig. 2.



CRH-neuron silencing after stress. Füzesi et al. (2016).

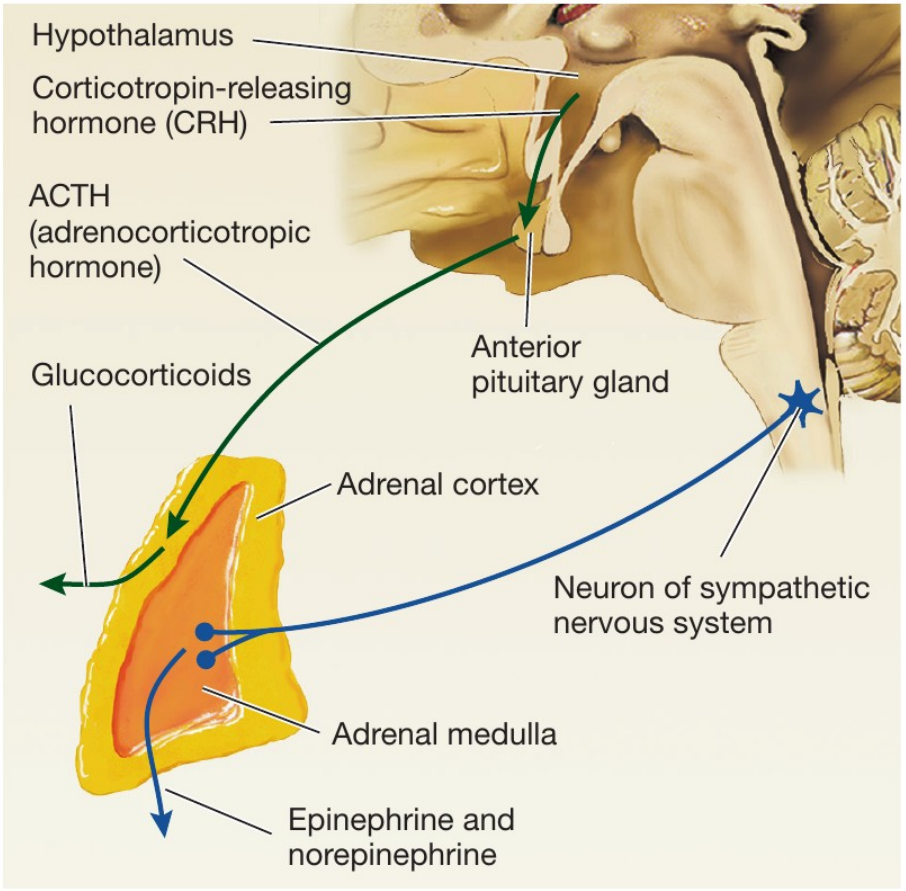
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The HPA axis carries a hormonal signal

The hypothalamic-pituitary-adrenal axis, or HPA axis, transfers information through three structures. Hypothalamic neurons release CRH into vessels serving the anterior pituitary. Pituitary cells respond by releasing adrenocorticotropic hormone, abbreviated ACTH, into the general circulation.

ACTH stimulates the adrenal cortex, the outer region of the adrenal gland. The cortex releases glucocorticoids, including cortisol in humans and mainly corticosterone in common laboratory rodents. These hormones change energy availability and influence cells in the immune system and brain.

Glucocorticoid receptors regulate cellular responses after the hormone enters a target cell. Some effects involve changes in gene expression and develop more slowly than a nerve impulse. Hormonal output and immediate autonomic output can therefore follow different time courses during the same stressful experience.



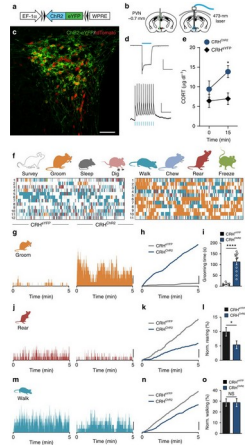
Carlson and Birkett (2022), Fig. 18.1, p.582.

Activating CRH neurons can initiate grooming

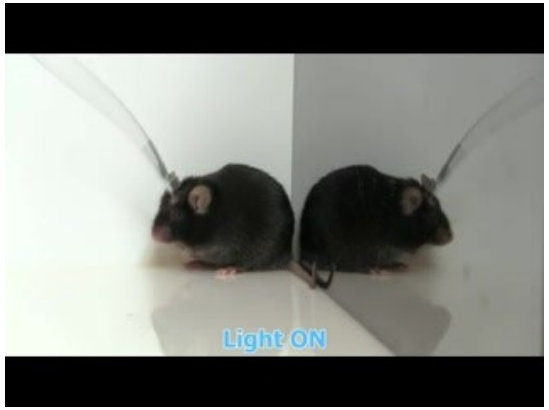
The mouse study also increased CRH-neuron activity with a light-sensitive excitatory protein. Excitation made grooming emerge even without a preceding stressor. The manipulation tested whether activity in this defined population could help generate an action within the post-stress repertoire.

Grooming consists of patterned movements directed toward the animal's own body. Scoring its onset and duration makes the action distinct from running or freezing. The original footage follows grooming during stimulation, rather than displaying a measurement of cortisol concentration or pituitary secretion.

Because these neurons can influence neural circuits as well as endocrine output, their behavioral effects need not be explained solely by circulating glucocorticoids. The experiment identifies a hypothalamic population that can organize an observable action; additional pathway tests are needed to assign that action to a particular output.



Füzesi et al. (2016), Fig. 3.



Grooming during CRH-neuron stimulation. Füzesi et al. (2016).

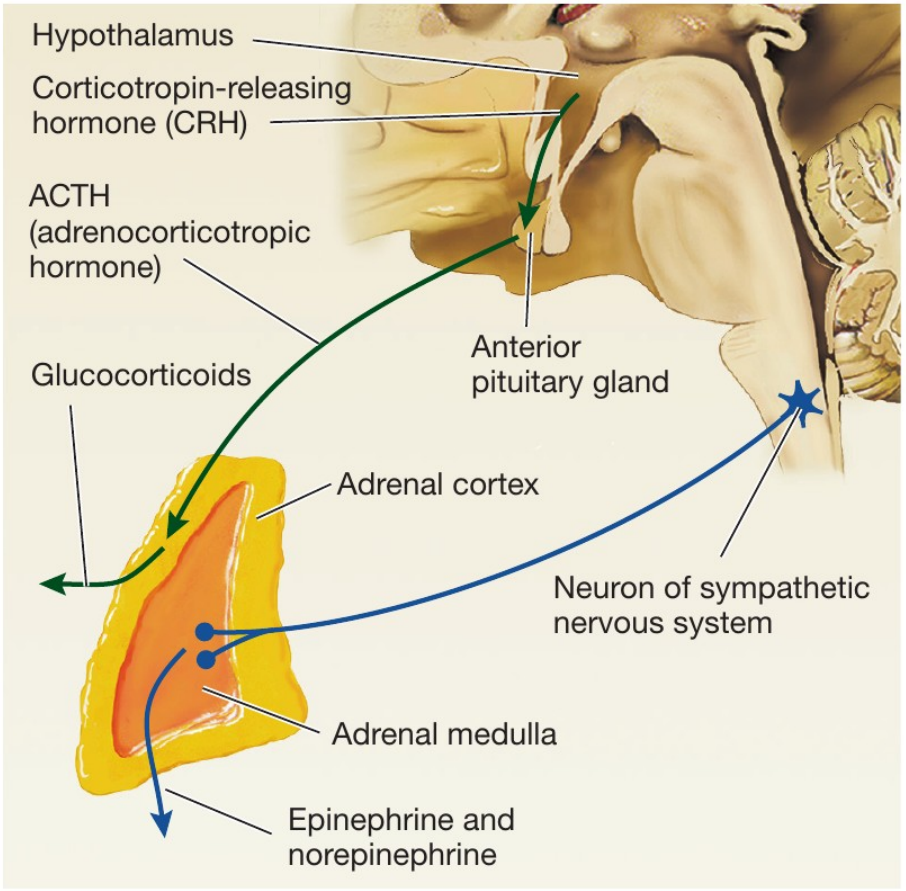
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Negative feedback limits glucocorticoid exposure

Glucocorticoids act back on the pituitary, hypothalamus, and other brain regions that regulate stress responses. This negative feedback reduces further drive through the HPA axis. A rise in the final hormone can therefore help turn down the system that produced it.

The hippocampus participates in contextual regulation of stress and contains receptors sensitive to glucocorticoids. Receptor occupancy and the duration of exposure influence how cells respond. A short hormone increase and repeatedly elevated exposure do not impose identical demands on these regulatory systems.

When challenges recur, recovery between episodes becomes important. Persistent activation can alter sleep, metabolism, and immune regulation even when an immediate response was useful. These longer effects concern the amount and timing of exposure, rather than a simple rule that any glucocorticoid release is harmful.



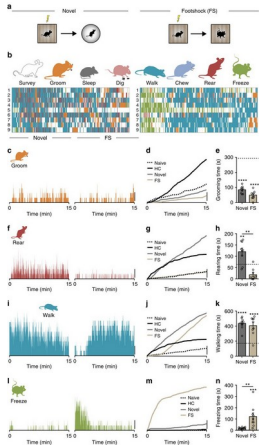
Carlson and Birkett (2022), Fig. 18.1, p.582.

A new environment changes the post-stress repertoire

Füzesi and colleagues placed stressed mice in different environments and scored their subsequent actions. In a novel environment, exploration competed with self-directed behaviors such as grooming. The animal's recent challenge therefore did not specify its next action independently of its surroundings.

The original recording follows a stressed mouse in the novel setting. Movement through the space provides an observation of exploration, while grooming is scored separately. These categories distinguish attending to an unfamiliar environment from repeating body-care movements in a familiar one.

The investigators also activated CRH neurons while changing the environmental setting. Excess activity reduced the normal sensitivity of the behavioral pattern to context. This result assigns a role to the hypothalamic cells in adjusting the repertoire, rather than merely detecting whether stress had occurred.



Füzesi et al. (2016), Fig. 5.



Post-stress behavior in a novel setting. Füzesi et al. (2016).

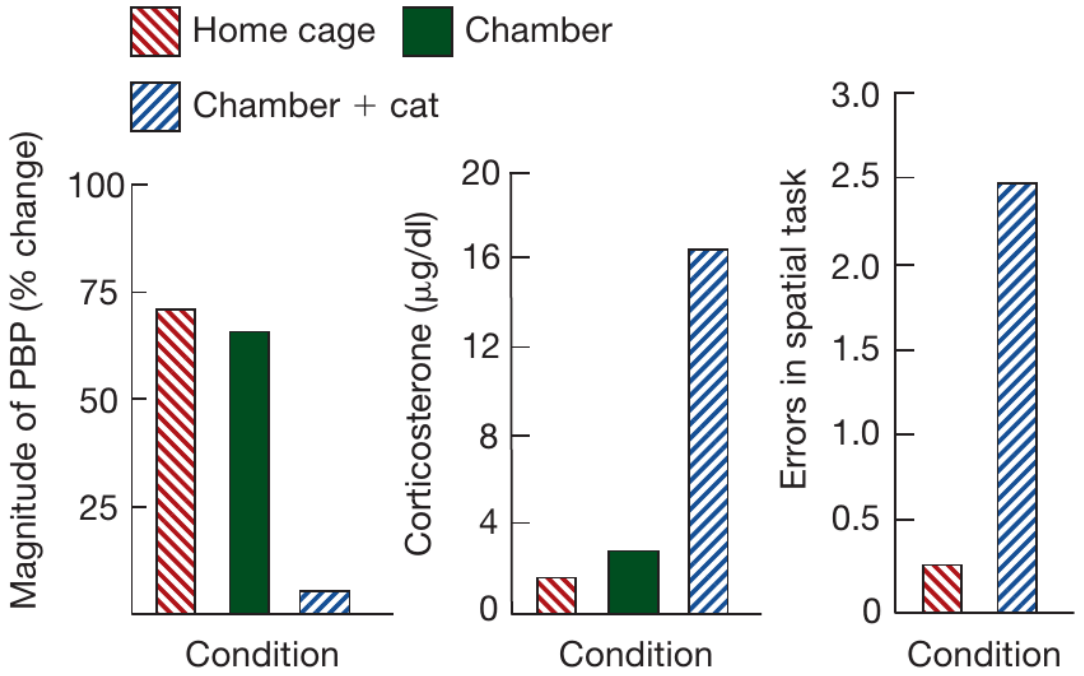
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Repeated stress can alter neural structure

Dendrites are the branches through which neurons receive much of their synaptic input. Prolonged stress can change dendritic branching and spine organization in regions such as the hippocampus and prefrontal cortex. These structural effects depend on the region, exposure pattern, and biological state of the animal.

The amygdala participates in evaluating threats and organizing defensive responses. Stress-related changes in this system need not follow the same direction as changes in hippocampal cells. A global description such as brain damage would hide these differences among circuits and cellular responses.

Altered plasticity can affect memory and regulation of later stress responses. Sleep, metabolic conditions, and opportunities for recovery also influence these processes. A structural change observed after repeated exposure is a different outcome from the immediate increase in heart rate during one challenge.



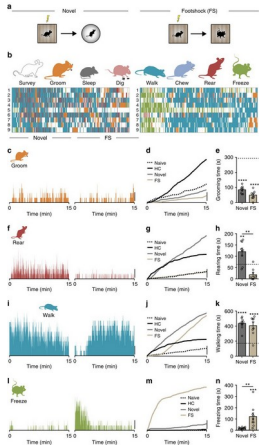
Carlson and Birkett (2022), Fig. 18.4, p.585.

Safety changes which actions follow stress

The CRH-neuron study compared behavior in settings that differed in perceived safety. An unsafe setting changed the distribution of actions after the stressor. The same recent event could therefore be followed by different behavior when the surrounding environment required continued vigilance.

In the original unsafe-environment recording, the mouse's movement and self-directed actions are observed after stress. These actions are interpreted within the actual setting, rather than labeled as evidence of a human disorder. Their timing shows how environmental demands compete with the usual post-stress sequence.

Optical activation of CRH cells made behavior less responsive to these contextual differences. Persistent drive in this population could therefore sustain a pattern that no longer matches current conditions. The experimentally measured result is reduced environmental adjustment of mouse behavior, not a clinical diagnosis.



Füzesi et al. (2016), Fig. 5.



Post-stress behavior in an unsafe setting. Füzesi et al. (2016).

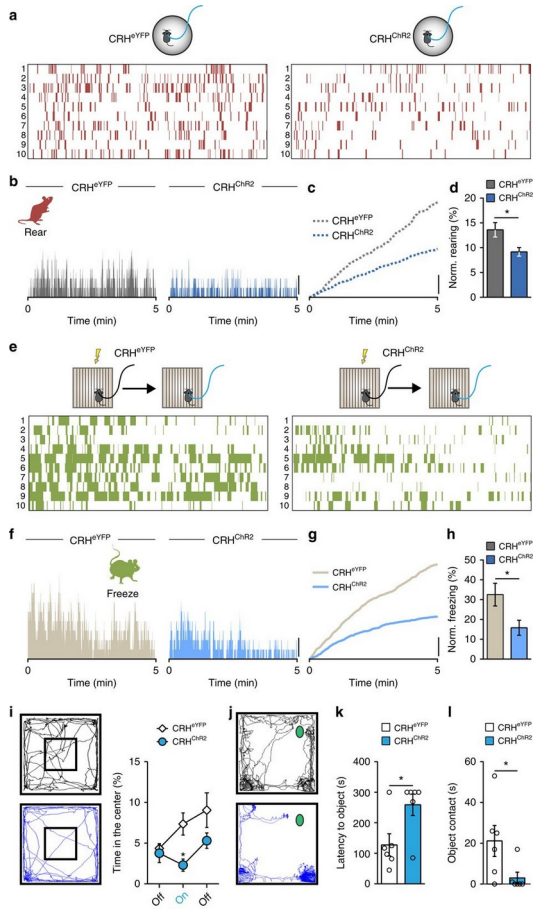
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CRH-cell output organizes a repertoire of actions

A behavioral repertoire is the set of actions available to an animal in a particular situation. The CRH experiments measured transitions among several actions rather than treating grooming as the only endpoint. This captured changes in both action selection and the organization of the session.

Inhibiting the cells after stress disturbed the expected repertoire, while exciting them could produce components of it without stress. These opposite manipulations assign a causal contribution to the cells. The behavioral effect is more specific than an association between hormone concentration and a later action.

Environmental comparisons further separated initiating an action from adjusting it to the setting. CRH-cell activation reduced this adjustment across contexts. A hypothalamic stress population can therefore influence both what an animal does and how strongly the surroundings modify that behavior.



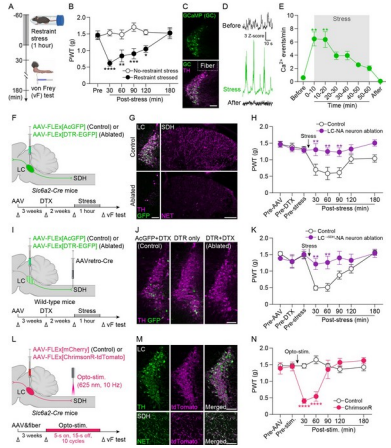
Füzesi et al. (2016), Fig. 7.

Stress recruits a descending pain pathway

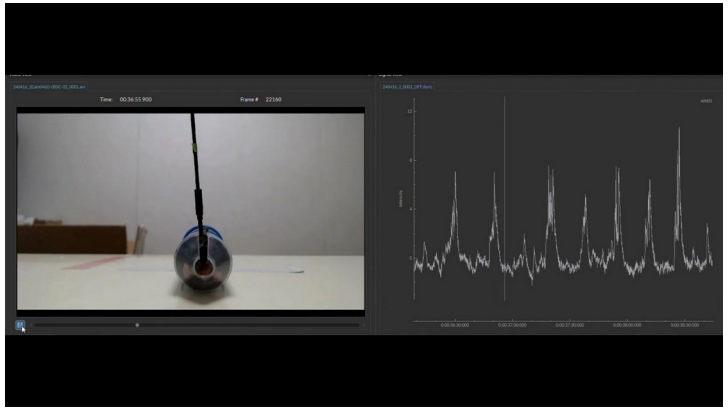
The locus coeruleus is a brainstem group of neurons that releases norepinephrine. A recent mouse study examined its descending projection to the spinal dorsal horn, where sensory information enters spinal circuits. This pathway can change how strongly mechanical stimulation produces a pain-related response.

During restraint stress, the investigators recorded increased activity in locus-coeruleus neurons. The accompanying recording aligns the restrained animal with a population fluorescence signal. Fiber photometry measures light from a group of labeled cells, rather than resolving the spikes of every individual neuron.

The study then tested spinal targets of the descending signal. Norepinephrine acted on a subset of astrocytes, the non-neuronal cells that regulate conditions around synapses. This provided a route by which a brain stress response could alter local sensory processing in the spinal cord.



Kawanabe-Kobayashi et al. (2026), Fig. 1.



Restraint and locus-coeruleus photometry. Kawanabe-Kobayashi et al. (2026).

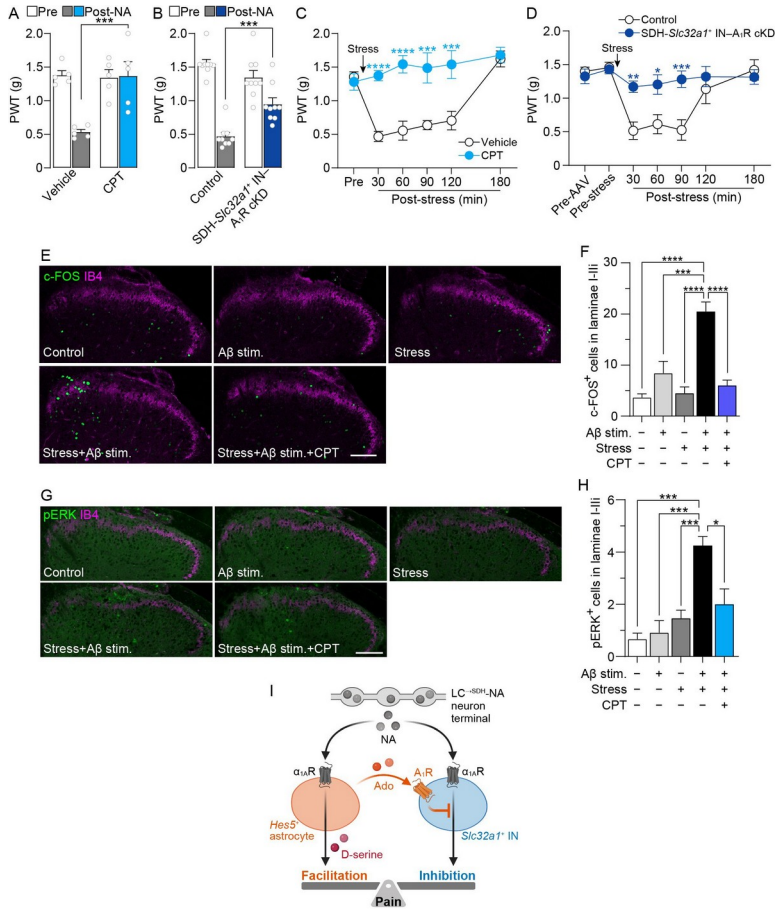
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Astrocytes can reduce spinal inhibition after stress

The spinal astrocytes examined in this study expressed alpha-1A adrenergic receptors. Norepinephrine activated these receptors and changed astrocyte signaling. The relevant cells belonged to a Hes5-labeled subset, rather than every astrocyte in the spinal cord responding identically.

Astrocyte signaling reduced the activity of inhibitory neurons through adenosine A1 receptors. Inhibitory neurons normally restrain other cells' activity, so reducing their output removes part of that restraint. This disinhibition increased the responsiveness of the sensory-processing circuit.

Reducing the relevant adenosine-receptor pathway prevented the stress-related increase in mechanical pain sensitivity. The intervention interrupted a defined step between astrocyte activity and neuronal inhibition. The measured effect concerned stress-induced hypersensitivity, not a claim that norepinephrine always increases pain in every pathway.



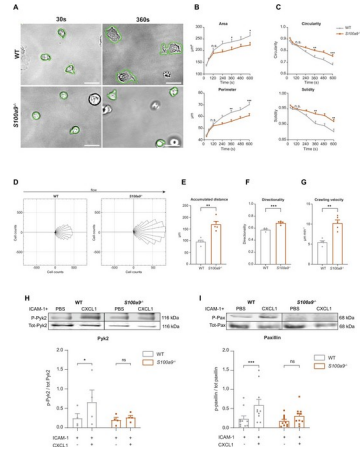
Kawanabe-Kobayashi et al. (2026), Fig. 4.

Neutrophils must attach before entering tissue

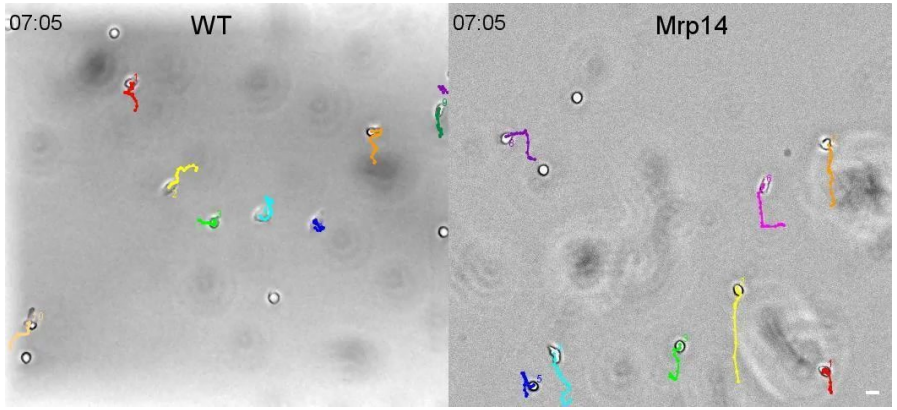
Neutrophils are immune cells that move from circulating blood toward inflamed tissue. Before crossing a vessel wall, they slow down, attach, spread, and crawl. A study of mouse neutrophils examined the cellular machinery that keeps attachment stable while fluid flow pushes against the cells.

The researchers compared ordinary cells with cells lacking S100A9, part of the calcium-binding S100A8/A9 protein complex. In a flow chamber, the deficient cells showed impaired polarization and crawling. Polarization means organizing a leading end and a trailing end for directed movement.

The original footage follows neutrophils on surfaces carrying adhesion molecules and a chemical guidance signal. This is an inflammatory-cell preparation, not a recording from psychologically stressed animals. It reveals a cellular process that stress hormones can regulate indirectly through changes in immune function.



Napoli et al. (2024), Fig. 3.



Neutrophil crawling under flow. Napoli et al. (2024).

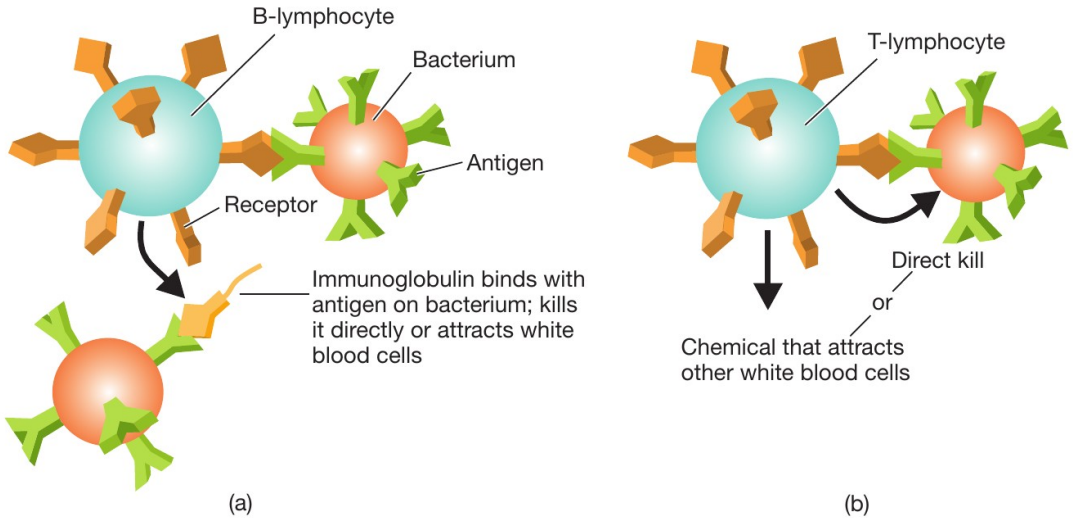
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Stress hormones change immune-cell activity

The immune system detects pathogens and damaged tissue through several kinds of cells. Innate immune cells, including neutrophils, respond rapidly to broad danger signals. Adaptive responses develop more specific recognition and can retain memory of earlier encounters with an antigen.

B cells can develop into plasma cells that secrete antibodies. T cells recognize antigen through their own receptors and can help coordinate responses or kill infected cells. Antibody production is therefore a B-cell function, even though helper T cells often support its development.

Glucocorticoids influence cell movement and the production of inflammatory signals. Their effects depend on timing, dose, and the state of the immune response. Persistent stress can disrupt protective regulation without implying that every immune cell is simply switched off at the same time.



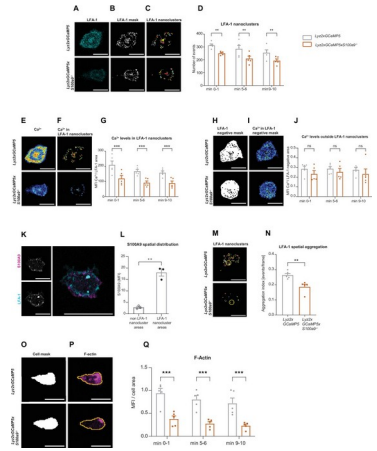
Carlson and Birkett (2022), Fig. 18.9, p.588.

Adhesion receptors gather at contact sites

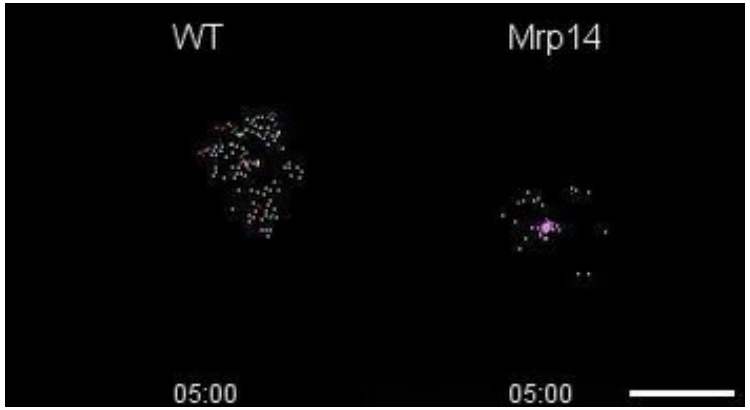
LFA-1 is an integrin, a receptor that helps a neutrophil attach to another surface. Binding to ICAM-1 helps create a stable contact with a vessel wall. Clusters of these receptors also transmit information from the contact site into the cell's interior.

The S100A9 study imaged LFA-1 clusters while neutrophils spread under flow. Cells lacking S100A9 had impaired post-attachment responses even though attachment machinery was present. The defect concerned organizing and sustaining the contact after the cell had begun to stop.

The original recording follows the changing clusters in ordinary and deficient neutrophils. A cluster is a local concentration of receptors, not a newly formed immune cell. Its position identifies where the cell receives adhesion-related signals and must coordinate its internal framework with the external surface.



Napoli et al. (2024), Fig. 4.



LFA-1 adhesion clusters during spreading. Napoli et al. (2024).

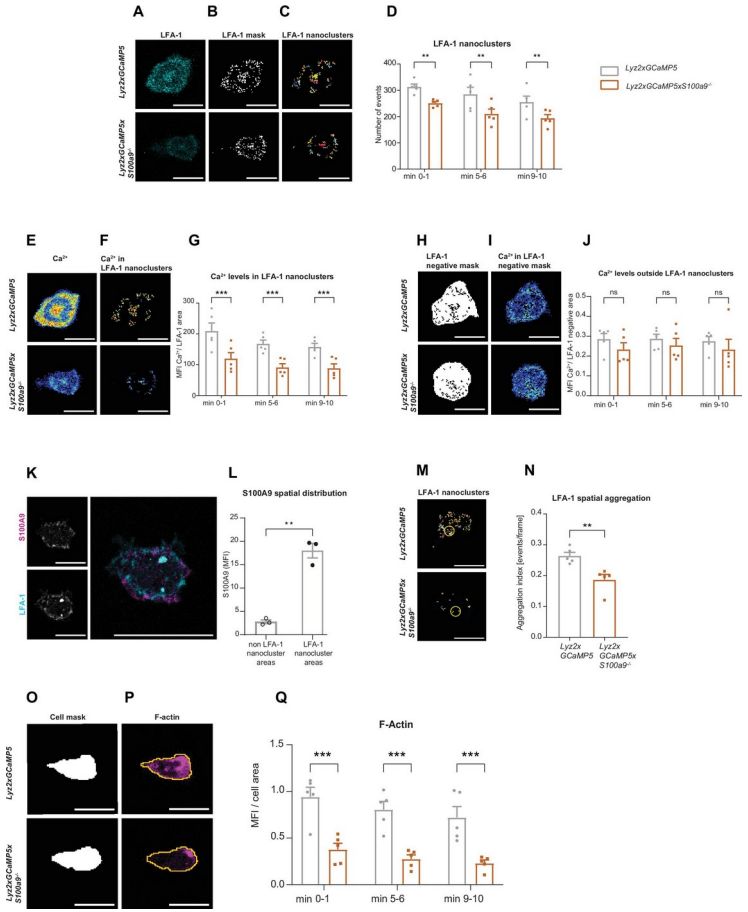
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Local calcium helps stabilize immune-cell attachment

Calcium ions act as signals inside neutrophils as well as neurons. A rise near an adhesion site can regulate proteins that organize cell shape. The location of that rise matters because spreading requires coordinated changes at the places where the cell contacts its surroundings.

The study measured reduced calcium availability at LFA-1 and actin clusters in S100A9-deficient neutrophils. Actin is a structural protein that forms filaments inside the cell. The calcium defect therefore occurred at sites that combine surface attachment with internal force and shape control.

The researchers separated local measurements from signals averaged across the whole cell. An altered average cannot identify where a useful calcium rise is missing. Imaging the contact regions revealed how a calcium-binding protein helps organize the spatial pattern required for stable recruitment into tissue.



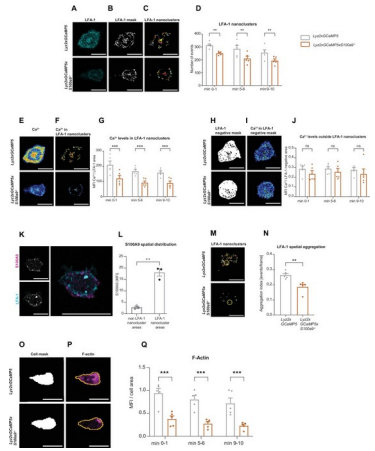
Napoli et al. (2024), Fig. 4.

Calcium signals differ at adhesion clusters

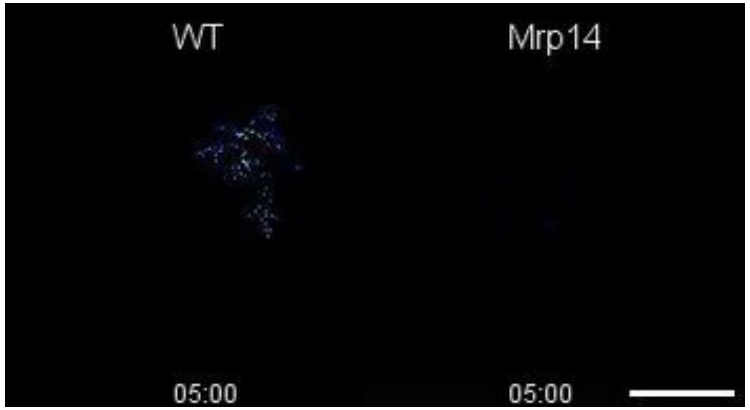
The neutrophil recordings compare calcium fluorescence at identified LFA-1 contact regions. Fluorescence increases when the calcium indicator binds calcium, allowing changes to be followed over time. The signal reports local calcium availability rather than the mechanical strength of the bond itself.

Deleting S100A9 disturbed these local calcium signatures during post-attachment responses. The cells consequently showed weaker spreading and polarization. The study connected the missing intracellular protein to a specific signaling defect before interpreting the later failure to remain attached under flow.

The original analysis uses masks to identify the receptor-cluster regions being measured. A mask marks selected pixels for measurement; it does not create a biological structure. Keeping the raw cell behavior and the identified regions distinct makes the spatial comparison understandable.



Napoli et al. (2024), Fig. 4.



Calcium at neutrophil adhesion regions. Napoli et al. (2024).

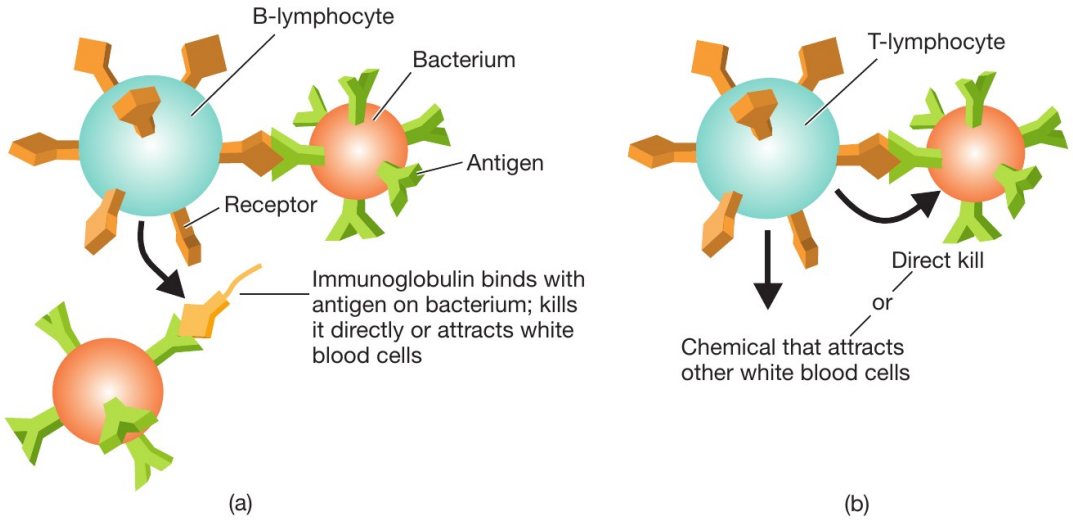
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Immune signals can influence brain and behavior

Cytokines are proteins through which immune cells influence other cells. During infection, inflammatory signals can change temperature, appetite, sleep, and activity. These coordinated sickness responses conserve resources and help organize the body's response to an ongoing biological challenge.

Information reaches the brain through several routes, including neural pathways and signaling at blood-brain interfaces. The effect is not explained by assuming that every circulating cytokine freely enters all brain tissue. Cells at these interfaces can translate a peripheral inflammatory signal into a local response.

Stress systems and immune systems regulate one another. The brain can alter endocrine and autonomic output, while inflammatory signals can recruit stress-related pathways. Disturbance can therefore develop through repeated interactions among systems rather than through one permanent direction of communication.



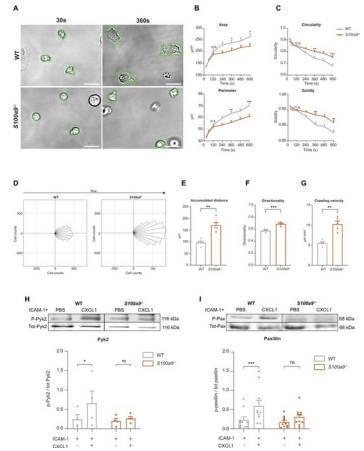
Carlson and Birkett (2022), Fig. 18.9, p.588.

Actin reorganization supports neutrophil spreading

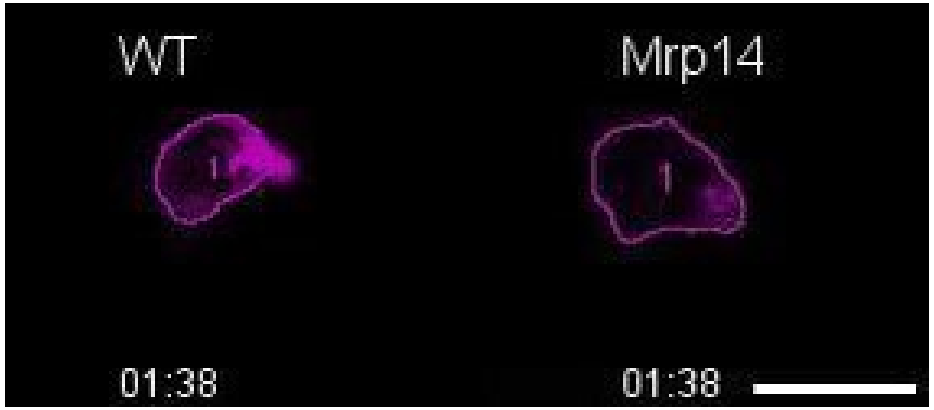
Actin filaments form part of the cytoskeleton, the internal framework that helps a cell change shape. A crawling neutrophil must build protrusions and organize force around its contacts. Stable recruitment requires this changing framework to cooperate with adhesion receptors at the surface.

S100A9-deficient cells showed impaired actin organization as they responded under flow. The original recording tracks the actin-defined cell region while cells attempt to spread. Reduced local calcium at contact clusters accompanied the difficulty in producing the usual polarized form.

The deficient cells detached more readily under fluid shear. Shear is the force created as moving fluid passes a surface. This final recruitment defect followed failures in local signaling and shape control, explaining how an intracellular calcium-regulating protein affects whether a circulating cell remains at an inflamed site.



Napoli et al. (2024), Fig. 3.



Actin-defined neutrophil shape under flow. Napoli et al. (2024).

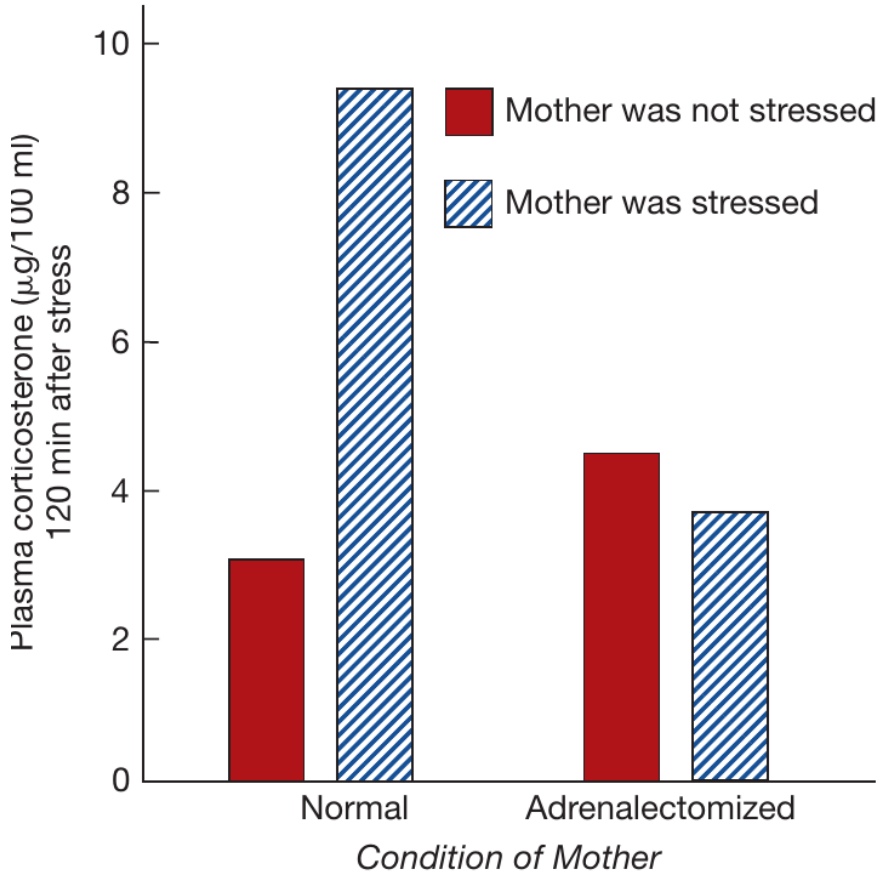
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Early experience can recalibrate stress regulation

Developing stress systems respond to the conditions in which an animal grows. Prenatal exposure and early caregiving can influence later HPA responses through changes in hormone exposure and neural regulation. Development therefore changes how future challenges are evaluated and how strongly they recruit physiological responses.

Rodent studies of maternal care have examined glucocorticoid-receptor expression in offspring. Receptor regulation can involve chemical modifications that affect gene activity without changing the DNA sequence. These epigenetic processes alter how accessible a gene is to the machinery that reads it.

Human development includes genetic differences and many interacting environments. An association between early adversity and later symptoms does not mean that every exposed child develops a disorder. Supportive relationships and later experience can also influence regulation, learning, and opportunities for recovery.



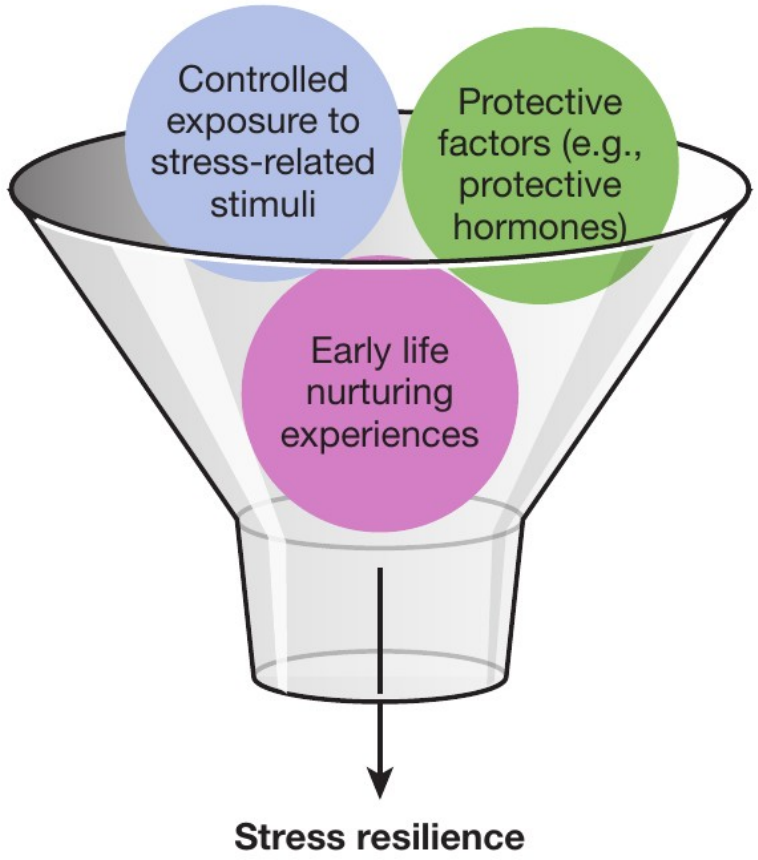
Carlson and Birkett (2022), Fig. 18.7, p.587.

Development changes vulnerability to later stress

The timing of adversity matters because developing circuits are still changing their connections and regulatory responses. Prenatal conditions can affect the fetus through maternal physiology and placental function. After birth, caregiving provides sensory experience and influences the infant's ability to recover from challenges.

Repeated early stress can alter later hormone responses, emotional learning, and attention to threat. These outcomes involve interactions between experience and the developing nervous system. A later response can therefore differ even when two individuals encounter a similar immediate stressor.

Resilience describes relatively preserved or recovered functioning despite adversity. It can reflect supportive environments, learned regulation, and biological differences rather than an absence of any physiological response. The outcome must be followed over time because initial distress and long-term disorder are different events.



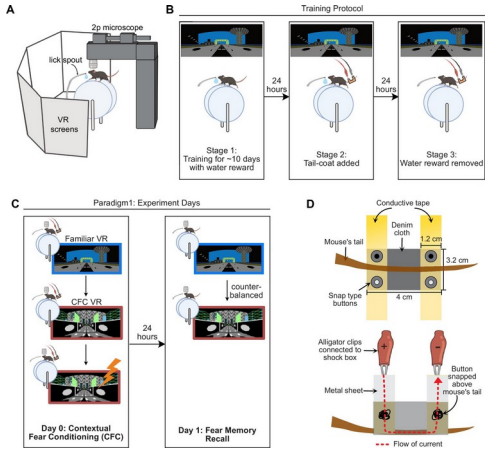
Carlson and Birkett (2022), Fig. 18.8, p.587.

A neutral context permits continued movement

A virtual-reality study examined how mice distinguish a neutral environment from one associated with an aversive event. Head-fixed mice moved through a visual setting while researchers tracked behavior and hippocampal activity. The apparatus allowed the same visual contexts to be presented repeatedly.

In the neutral-context recall recording, the mouse continued moving through the virtual environment. This provides a comparison for movement in the previously conditioned setting. The recall phase presents the learned context without delivering the original aversive event in the selected recording.

The hippocampus contains place cells, neurons whose activity varies with location. Recording these cells allowed the investigators to ask how contextual fear changes the representation of space. The measured representations belonged to mice performing the virtual task, rather than people recalling a traumatic experience.



Krishnan et al. (2025), Fig. 1.



Neutral virtual-context recall. Krishnan et al. (2025).

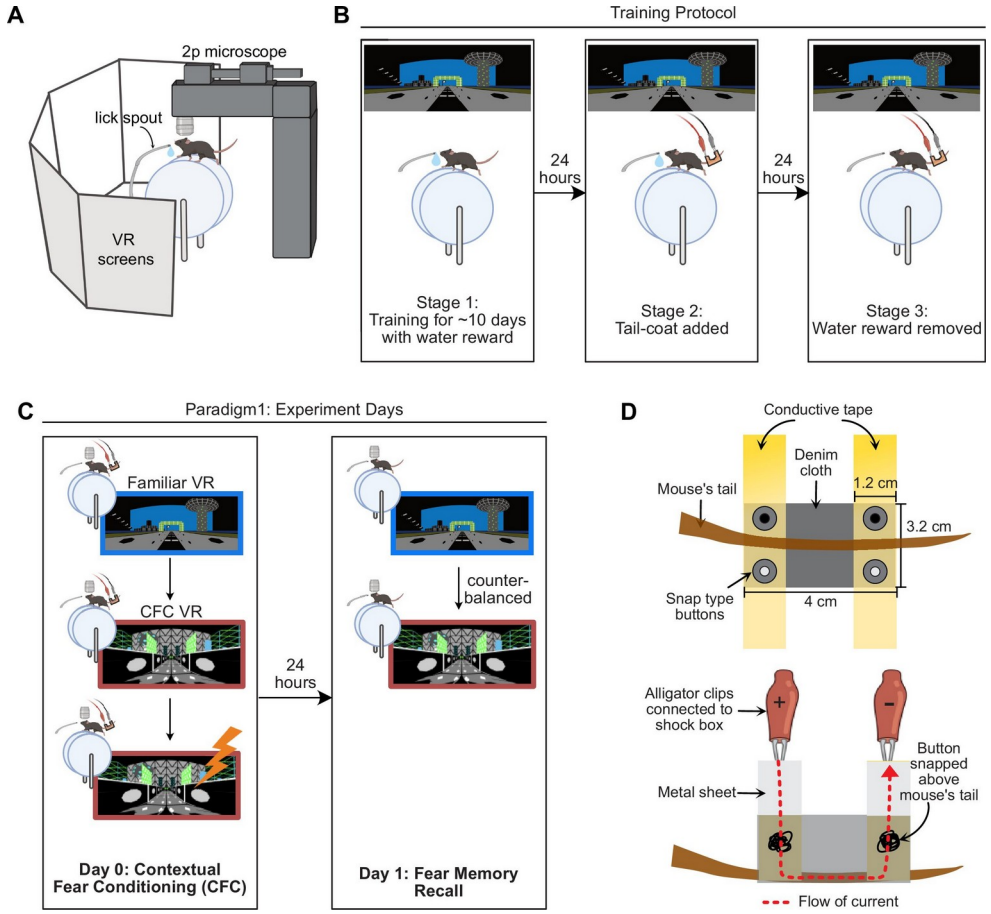
Play video

PTSD includes persistent changes after trauma

Posttraumatic stress disorder, or PTSD, can develop after exposure to a traumatic event. Symptoms include intrusive memories, distressing dreams, or experiences in which the event seems to recur. Reminders can trigger strong emotional and physiological responses even when the original danger is absent.

Avoidance can extend to places, conversations, thoughts, and other reminders. Persistent changes in mood and beliefs may include guilt, detachment, or reduced positive emotion. Heightened vigilance, exaggerated startle, sleep disturbance, and difficulty concentrating form another part of the symptom pattern.

Diagnosis depends on the pattern, persistence, and effects on functioning, rather than one fearful response. Many people experience distress after trauma without developing persistent PTSD. Trauma exposure establishes the relevant event, while subsequent symptoms and recovery describe the person's later course.



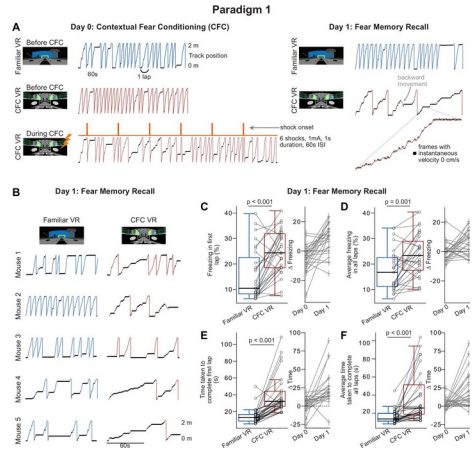
Krishnan et al. (2025), Fig. 1.

A learned context can suppress movement

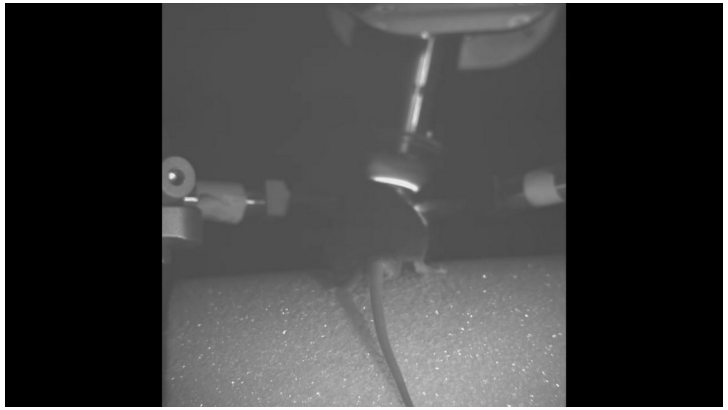
In the virtual-context experiment, mice learned that one visual environment predicted an aversive event. During later recall, entering that setting increased freezing. Freezing is a sustained reduction in movement used here as one observable defensive response, separate from subjective human fear.

The original recall recording follows reduced movement in the conditioned context. Researchers compared this behavior with the neutral context presented in the same apparatus. Because the selected recall sequence contains no new aversive event, the response depends on what the animal previously learned about the setting.

Hippocampal place-cell activity also changed after conditioning. The spatial representation was reorganized rather than remaining a fixed map independent of emotional experience. This permits contextual learning to alter which locations recruit activity during a later encounter with the environment.



Krishnan et al. (2025), Fig. 2.



Freezing in a conditioned virtual context. Krishnan et al. (2025).

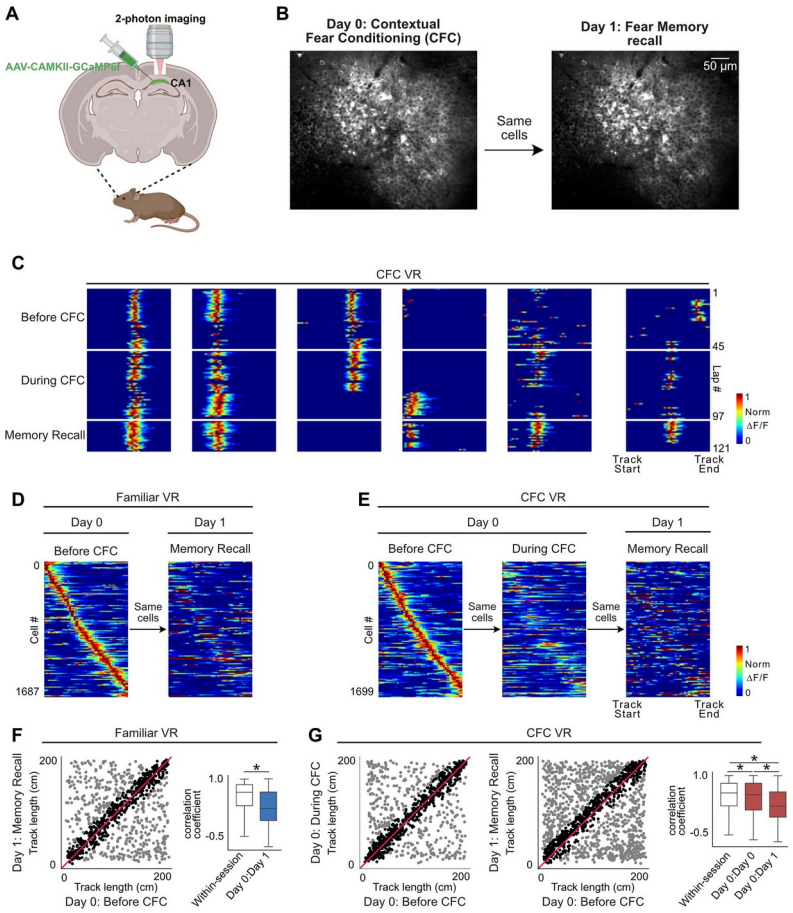
Play video

The hippocampus helps distinguish contexts

A context combines features such as location, visual surroundings, and the sequence of events. The hippocampus helps represent these combinations and retrieve memories associated with them. A danger signal encountered in one setting can therefore be interpreted differently in another setting.

Contextual discrimination reduces inappropriate generalization of a learned response. When distinct safe situations are treated as equivalent to the dangerous one, defensive behavior can spread beyond the original setting. This is different from learning that a single sound predicts an aversive event wherever it occurs.

The amygdala receives information relevant to threat and influences defensive outputs. Hippocampal information helps specify the circumstances in which those responses are appropriate. Communication among these regions provides a route through which memories of setting can regulate the expression of learned fear.



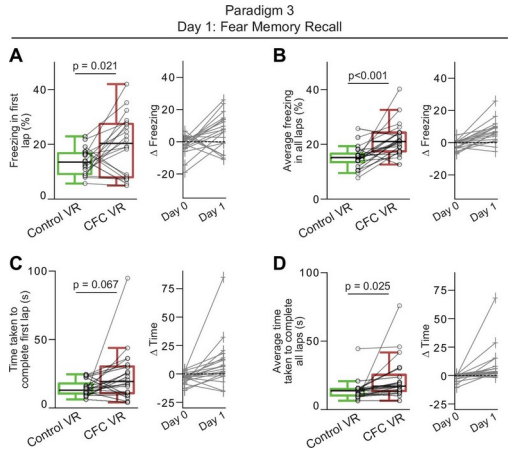
Krishnan et al. (2025), Fig. 7.

Contextual fear can produce avoidance as well as freezing

Not every conditioned mouse responded only by becoming still. The virtual-reality study also recorded avoidance of the learned aversive setting. Some animals moved backward when approaching its visual features, producing an active response rather than a simple absence of movement.

The selected recording follows this backward movement during recall. Direction is therefore necessary for interpreting the behavior: a reduction in forward progress could reflect retreat rather than freezing. Measuring only total movement would combine two different defensive strategies into one value.

The study related these behavioral differences to hippocampal representations of the environments. Changes in place-cell organization accompanied contextual fear learning. The neural data concern how space is represented during learned defense, while the behavior identifies whether the animal stops or actively withdraws.



Krishnan et al. (2025), Fig. 6.



Retreat in the conditioned virtual context. Krishnan et al. (2025).

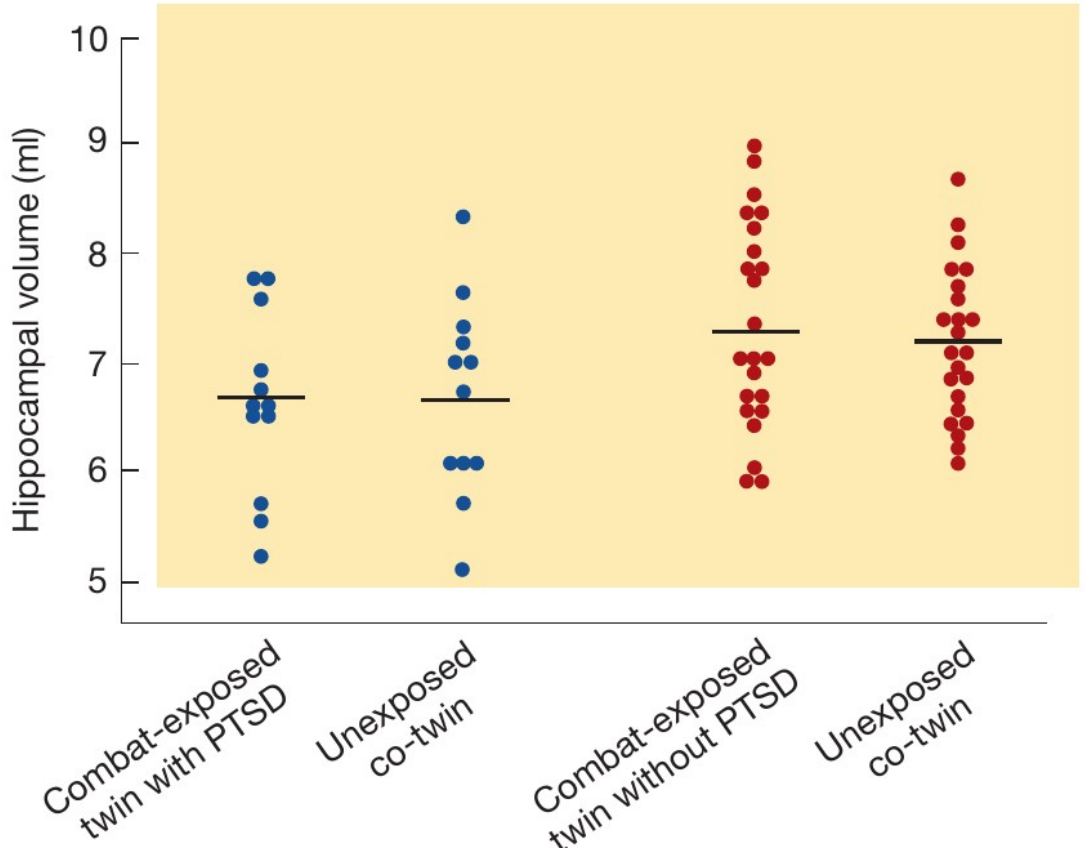
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A smaller hippocampus can precede trauma exposure

Imaging studies have reported smaller hippocampal volume in some groups with PTSD. A volume difference measured after trauma does not establish when it began. It could reflect pre-existing vulnerability, later experience, or processes that contribute at more than one stage.

A twin study compared combat-exposed men and their genetically identical brothers who had not experienced combat. Smaller hippocampal volumes were also found in the unexposed brothers of men with PTSD. That pattern suggested that part of the difference could precede the traumatic exposure.

The hippocampus participates in contextual memory, so a pre-existing difference could affect discrimination of safe and dangerous circumstances. Brain volume alone does not measure the accuracy of every memory process. The twin comparison addresses timing of vulnerability rather than diagnosing PTSD from an anatomical scan.



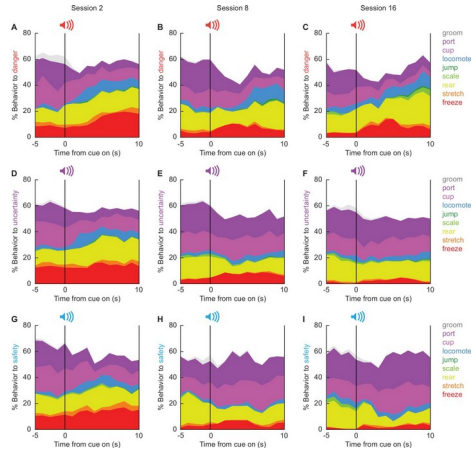
Carlson and Birkett (2022), Fig. 18.14, p.593.

A danger cue organizes several defensive actions

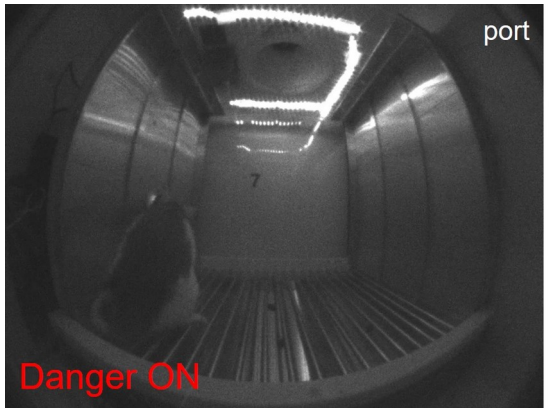
A rat study measured the sequence of behaviors elicited by a cue that predicted danger. The researchers scored freezing, rearing, locomotion, jumping, and suppression of reward seeking. The cue therefore produced a changing collection of actions rather than one uniform response throughout the trial.

The original female-rat recording includes observer labels aligned with the behavior. Rearing raises the body for investigation, while freezing reduces movement and port-related actions concern the reward apparatus. The labels make transitions visible without treating every interruption of reward seeking as freezing.

The timing of the predicted aversive event helped organize the sequence. Different portions of a cue could recruit different responses as danger became more imminent. This permits an animal to gather information or reposition before switching to a more immediate defensive action.



Chu et al. (2024), Fig. 3.



Danger-cue ethogram: female rat. Chu et al. (2024).

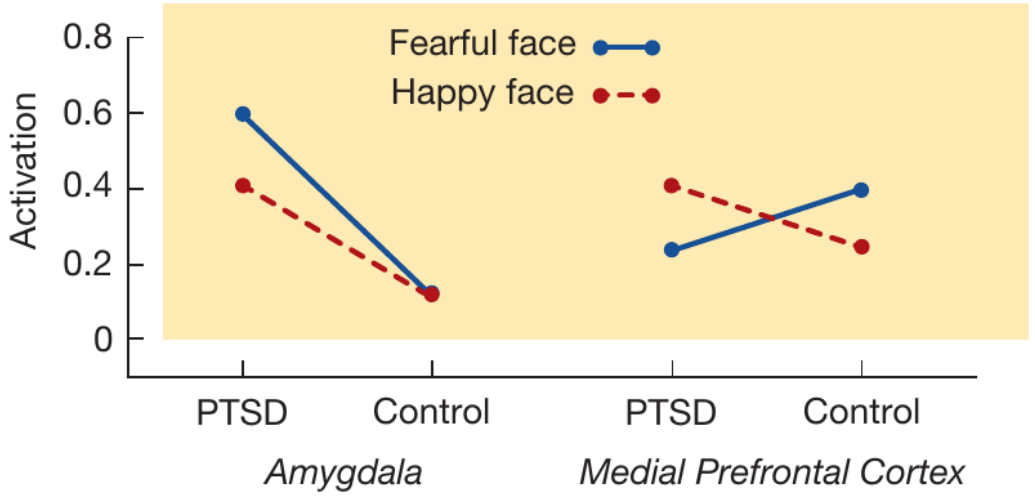
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Prefrontal regulation changes fear expression

The prefrontal cortex helps regulate behavior according to goals, context, and learned consequences. Its interactions with the amygdala can influence whether a defensive response is expressed. Regulation depends on activity in particular pathways rather than a single cortical switch that erases fear.

During extinction learning, a formerly threatening cue is presented without its expected aversive consequence. New learning can reduce the response while the original association remains retrievable. A return of fear in another context therefore does not necessarily mean that no extinction learning occurred.

In PTSD, persistent threat responses can involve altered coordination among prefrontal, hippocampal, and amygdala systems. These regions contribute different information about regulation, context, and significance. Their roles should be separated from the symptom descriptions used to establish the clinical diagnosis.



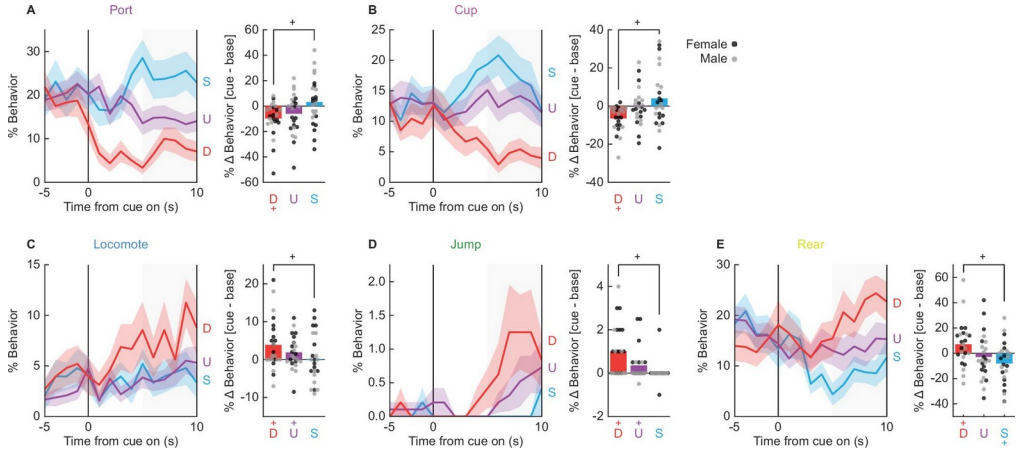
Carlson and Birkett (2022), Fig. 18.15, p.593.

The same danger cue allows different action sequences

The second rat recording follows a male animal responding to the danger cue. Its scored actions again change across the trial rather than staying in one defensive state. Comparing the two examples reveals that an identical task structure can permit different combinations of observable behavior.

The study quantified these actions across animals and time, including the interruption of reward seeking. A decrease in reward-port activity can occur because the animal freezes, investigates, or performs another action. Scoring the alternatives identifies what actually replaced the appetitive behavior.

The selected male and female examples are individual demonstrations, not evidence by themselves for a sex difference. Their teaching value is the visible sequence of actions within the conditioned trial. The paper's group analyses, rather than the sex of two examples, establish its findings about defensive organization.



Chu et al. (2024), Fig. 4.



Danger-cue ethogram: male rat. Chu et al. (2024).

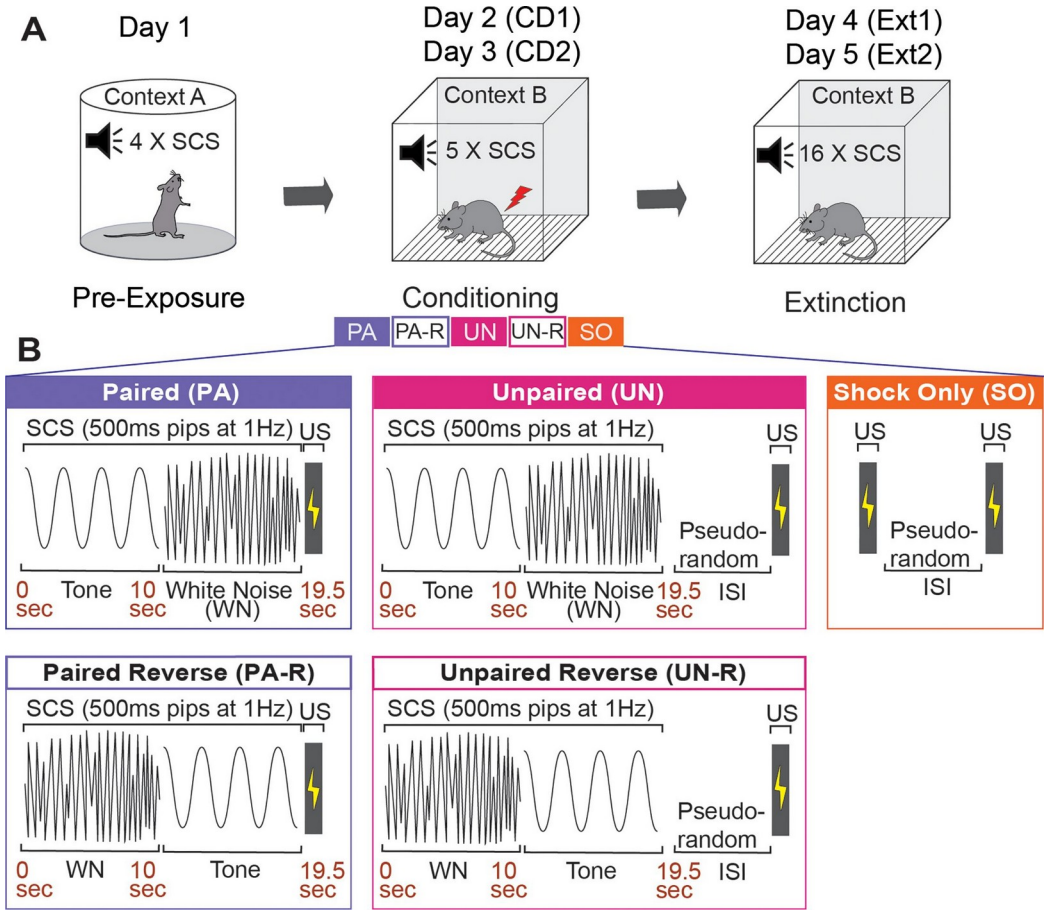
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An ethogram separates the actions being measured

An ethogram is a defined list of behaviors used to score an animal's activity. Each category requires an observable rule so that different observers can classify the same event. Freezing, rearing, escape, and grooming must therefore be distinguished before their frequencies or timing can be compared.

A single summary score can hide transitions among these actions. Two animals may spend the same total time away from a reward port while performing different behaviors. An ethogram preserves the difference between remaining still and actively moving away from a predicted threat.

Behavior also depends on stimulus order and prior experience. A loud sound can produce immediate movement through its sensory properties, while a previously neutral sound can acquire meaning through learning. Separate scoring helps identify which action reflects the learned prediction and which reflects an immediate reaction.



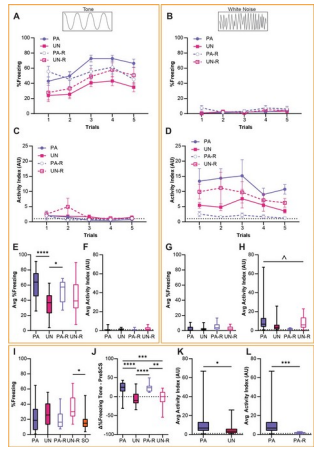
Le et al. (2024), Fig. 1.

Pairing changes how a sound sequence controls defense

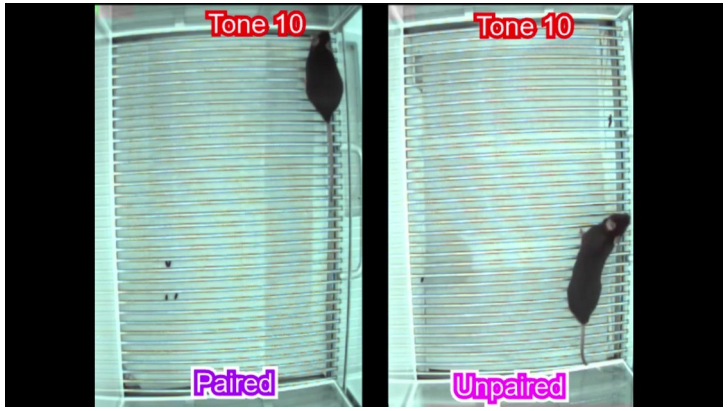
A mouse study presented a serial compound stimulus, a sequence containing a tone and white noise. Some mice experienced the sequence paired with an aversive event, while comparison groups received different arrangements. Pairing allowed the sounds to become predictors rather than merely repeated sensory stimulation.

The paired sequence produced organized freezing and flight-like responses to its components. The original comparison includes paired and unpaired conditions during conditioning. These conditions distinguish learning about the relationship between events from changes caused by exposure to the sounds or stress alone.

White noise has sensory properties that can provoke movement even without the same learned association as the tone. The investigators therefore measured multiple actions across both components. The resulting pattern cannot be described accurately by assigning every response in the sequence to one fear-memory process.



Le et al. (2024), Fig. 2.



Paired and unpaired sound-sequence conditioning. Le et al. (2024).

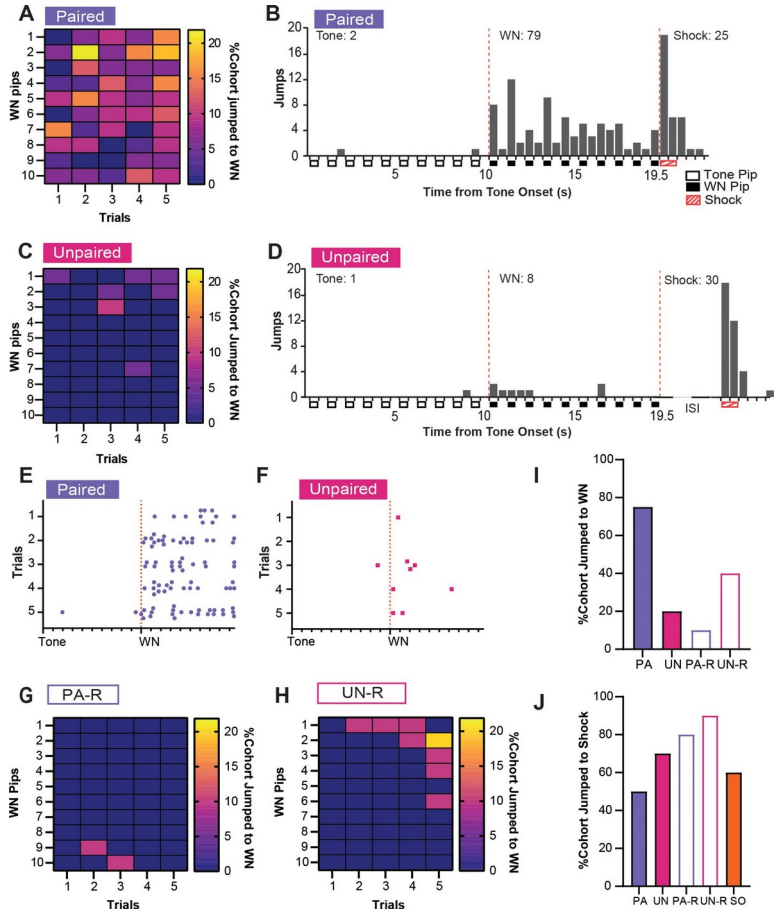
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Association and sensitization have different causes

Associative learning changes behavior because one event predicts another. Sensitization increases responsiveness after a strong experience without requiring that predictive relationship. The defensive-ethogram study arranged cue and aversive events differently to separate these sources of behavioral change.

Mice in paired, unpaired, and aversive-event-only conditions did not receive identical information about the sounds. A response shared across conditions could arise without the specific pairing. A response selectively strengthened by the predictive arrangement provides evidence for an associative component.

The tone and white-noise portions also differed in the actions they evoked. Measuring freezing alone would miss some of the active responses. Following the complete sequence made it possible to compare learned timing with immediate sensory effects in the same behavioral session.



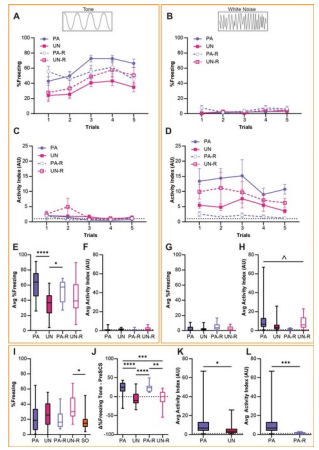
Le et al. (2024), Fig. 3.

Reversing the sound order changes the learned sequence

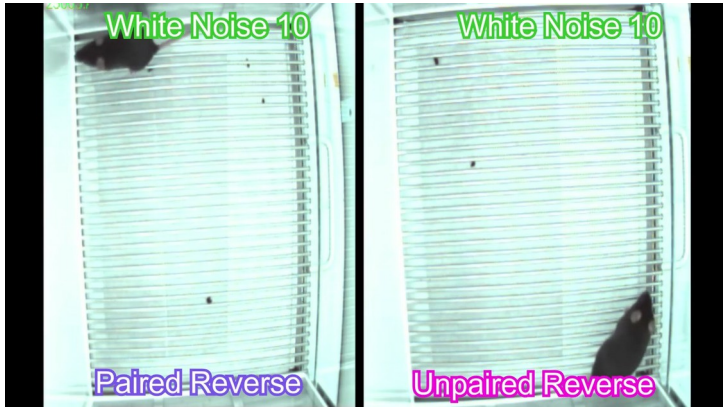
The investigators reversed the order of tone and white noise in a second set of conditioning arrangements. This changed which sensory event came earlier and which was nearer the aversive outcome. Temporal position therefore varied while the two basic sound types remained available.

The original reverse-order recording compares paired and unpaired mice during conditioning. Its visible movements show how the animals respond as the sequence unfolds. Reversing the order tests whether a particular action belongs inherently to a sound or depends partly on its learned temporal position.

The resulting defensive patterns differed from those trained with the original order. This indicates that the sequence is not simply a collection of independent reflexes. The animal learns relationships among successive events, and those relationships help determine when freezing or active movement occurs.



Le et al. (2024), Fig. 2.



Reverse-order sound-sequence conditioning. Le et al. (2024).

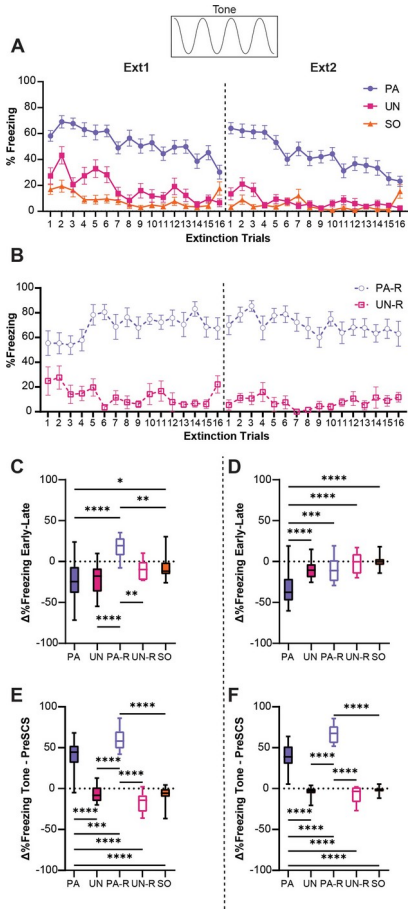
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Exposure builds new learning about safety

Exposure-based treatment repeatedly brings a person into contact with a feared cue or situation under conditions that permit new learning. The person can discover that the expected harmful outcome does not occur. Avoiding every reminder would prevent that corrective experience from taking place.

Extinction is the laboratory process in which a conditioned cue is presented without its predicted aversive event. It often reduces the learned response but does not necessarily delete the original memory. Context and later experience can influence which memory controls behavior during a future encounter.

Treatment therefore includes applying learning across useful situations rather than relying only on one familiar setting. The clinical goal concerns distress, avoidance, and functioning in people. Reduced freezing in a mouse extinction session measures one related learning process at a different biological and behavioral scale.



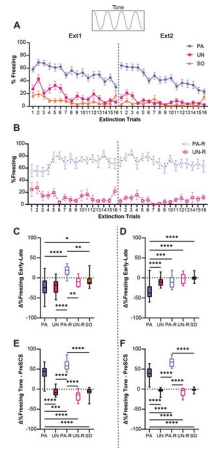
Le et al. (2024), Fig. 5.

Extinction changes responses within the learned sequence

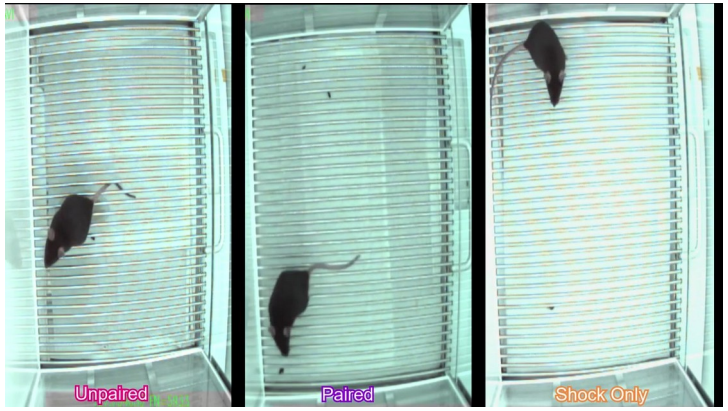
The defensive-ethogram study followed mice during early and late extinction trials. The sound sequence was presented without the expected aversive outcome. The investigators compared paired, unpaired, and aversive-event-only groups to identify how earlier training affected the later response.

The original recording places early and late trials beside one another. Tone-related freezing decreased during extinction, while the full ethogram preserved information about movement during the other component. A reduction in one action did not require every behavior to change in exactly the same way.

These comparisons distinguish the immediate response to a sound from the learned expectation attached to it. Repeated safe presentations can alter the expression of the association. The preceding conditioning arrangement still matters because it determines what prediction the animal must revise during extinction.



Le et al. (2024), Fig. 5.



Early and late extinction comparisons. Le et al. (2024).

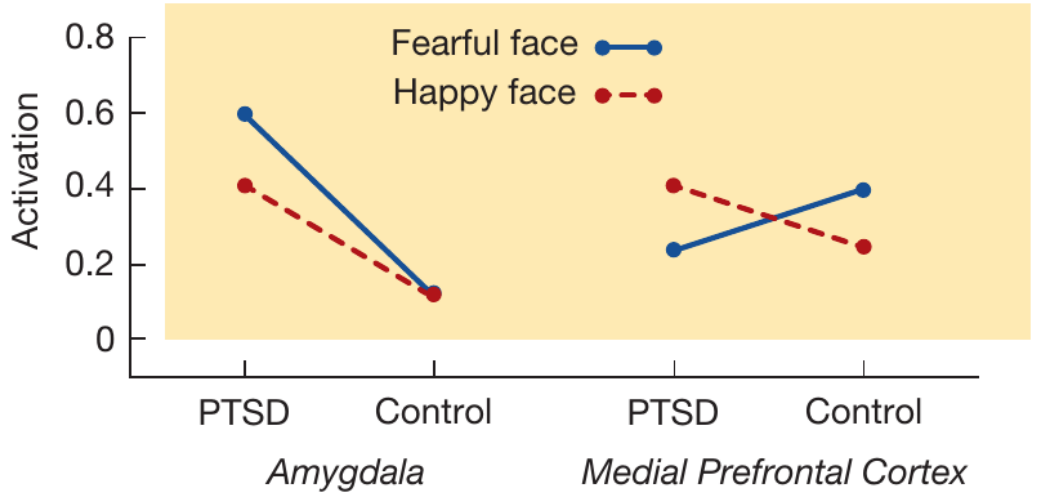
Play video

PTSD treatment targets symptoms and functioning

Trauma-focused psychological treatments help people work with memories, interpretations, and avoidance related to trauma. Exposure-based approaches provide structured opportunities for new learning. Treatment is adapted to the person rather than assuming that every distressing reminder should be confronted in the same way.

Some antidepressant medicines reduce PTSD symptoms through changes in monoamine signaling and later adaptation. Selective serotonin reuptake inhibitors, or SSRIs, block the serotonin transporter. They slow transmitter removal from the space outside neurons rather than directly acting as serotonin-receptor agonists.

Brain-stimulation approaches are also being investigated for selected symptoms and treatment-resistant conditions. Transcranial magnetic stimulation changes neural activity through magnetic pulses applied outside the head. A proposed circuit target and a demonstrated clinical improvement are separate findings that require their own measurements.



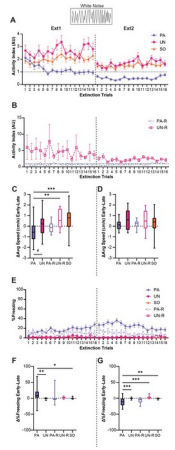
Carlson and Birkett (2022), Fig. 18.15, p.593.

Training order can change persistence during extinction

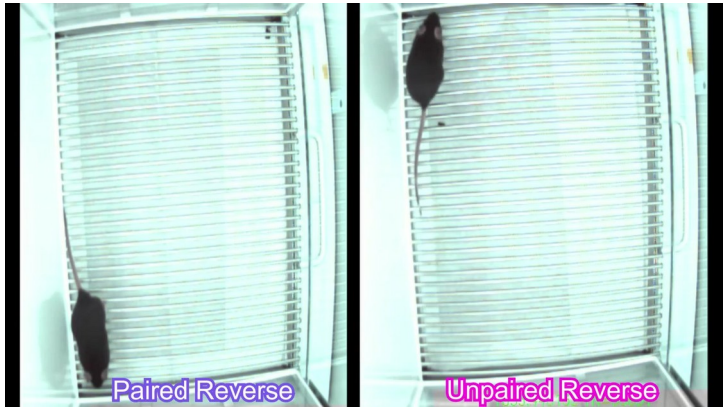
The reverse-order groups also received extinction sessions. In the paired reverse arrangement, tone-related freezing was relatively resistant to extinction. Changing the original temporal arrangement therefore changed how readily the later response declined when the outcome stopped occurring.

The selected recording compares early and late extinction in reverse paired and unpaired groups. The labels identify the original arrangement before the sound sequence begins. The comparison follows the animal's learned history rather than assuming that one sound always has a fixed emotional meaning.

A persistent response can therefore arise from the structure of prior learning, not simply from greater movement or a stronger immediate reaction to noise. The experiment identifies a condition that changes extinction of a defined response. It does not establish that every persistent human symptom has the same cause.



Le et al. (2024), Fig. 6.



Reverse-order extinction comparisons. Le et al. (2024).

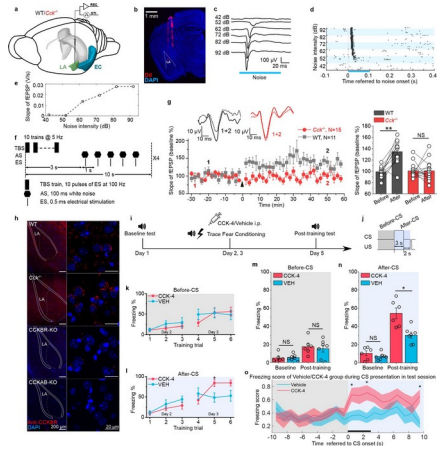
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CCK signaling helps connect separated events

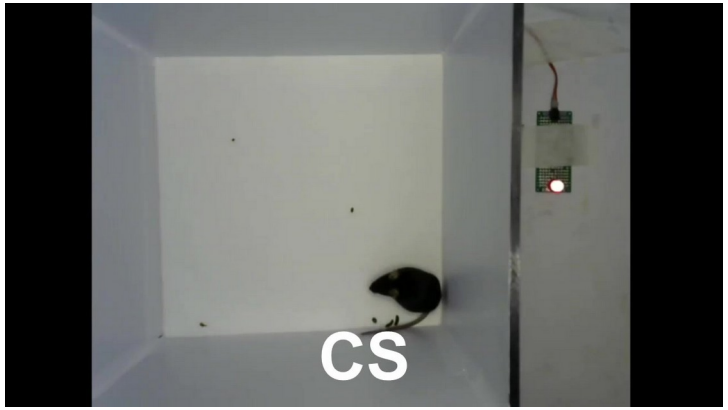
Trace fear conditioning places a silent interval between a sound and the later aversive event. Remembering the relationship requires retaining information across that gap. Researchers examined cholecystokinin, or CCK, a peptide messenger released by entorhinal-cortex neurons that project to the lateral amygdala.

In mice lacking CCK, the investigators found impaired trace fear learning. CCK-related signaling also contributed to strengthening auditory responses in the amygdala. Long-term potentiation means a lasting increase in the effectiveness of synaptic input, providing a cellular change through which the sound can acquire predictive value.

The selected recording follows a CCK-deficient mouse given CCK-4 during the short-trace experiment. Freezing during the later sound test demonstrates the recorded rescue condition. This is a mouse memory experiment; it is not a recording of a human panic attack or a clinical CCK treatment.



Feng et al. (2021), Fig. 2.



CCK-4 rescue condition in a Cck-knockout mouse. Feng et al. (2021).

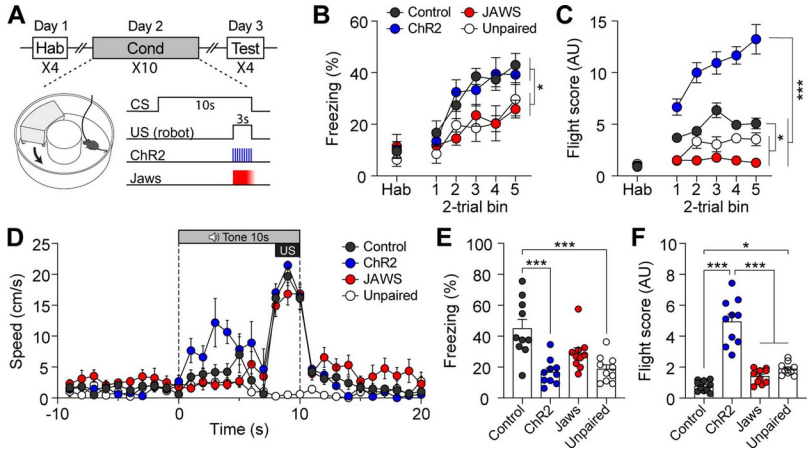
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An alarm population can strengthen active defense

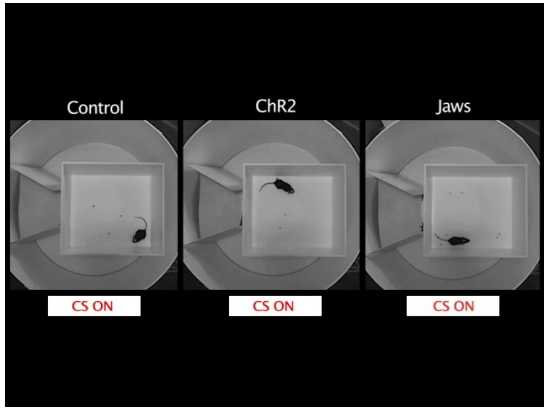
Researchers used a predator-like robot to study active defense in mice. They examined neurons containing calcitonin gene-related peptide, or CGRP, in the parabrachial nucleus of the brainstem. These neurons respond to aversive information and send signals toward forebrain systems involved in defensive behavior.

Activating the CGRP population during robot-related conditioning increased flight responses, while reducing its activity weakened them. The experiment extended the cells' role beyond passive freezing. The measured action depended on a task in which escaping an approaching threat was possible.

The selected recording follows later recall, with comparison groups responding to the learned cue. The original robot is absent from this recall view. Greater flight therefore concerns expression of the learned defensive response, rather than the immediate mechanics of a robot physically chasing the animal in this clip.



Pyeon et al. (2025), Fig. 3.



Defensive recall after CGRP manipulation. Pyeon et al. (2025).

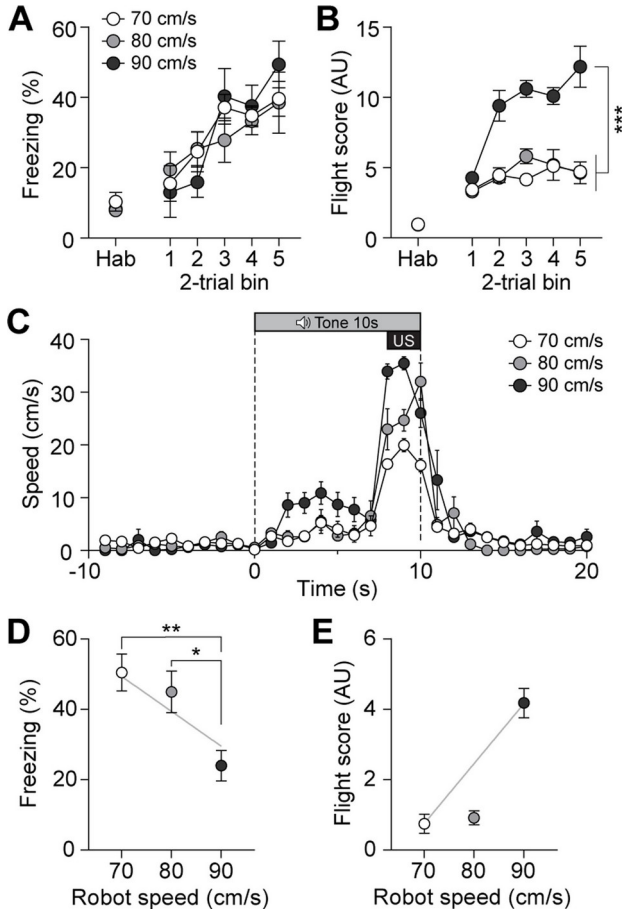
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Fear and anxiety differ in their time course

Fear commonly refers to responses to a relatively immediate or identifiable threat. Anxiety often involves anticipation of possible danger that is uncertain or more distant. Both can recruit attention, autonomic changes, and defensive behavior, but their triggers and persistence need not be identical.

A useful defensive response becomes clinically concerning when its intensity, duration, or effects on functioning are disproportionate to the circumstances. Occasional fear is not itself an anxiety disorder. Assessment considers the person's experience, avoidance, physiological symptoms, and ability to participate in daily activities.

Animal experiments isolate components such as threat detection, avoidance, or learned freezing. They can identify circuit mechanisms without reproducing the full human experience of worry or panic. A neural population that changes escape behavior therefore addresses one defined component of defensive regulation.



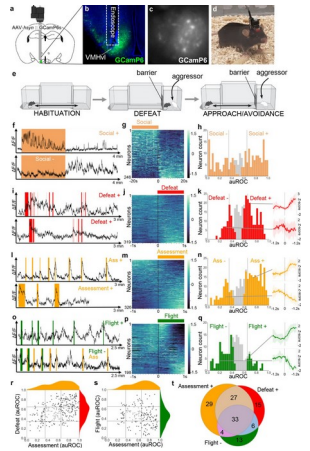
Pyeon et al. (2025), Fig. 5.

Social assessment recruits hypothalamic activity

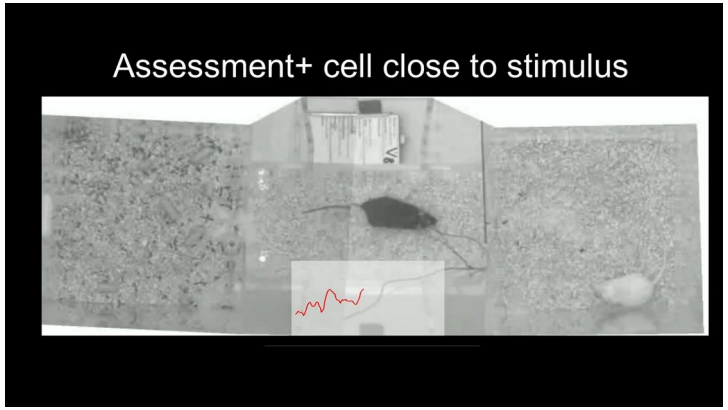
A mouse study examined neurons in the ventrolateral ventromedial hypothalamus during social threat. The investigators followed cells expressing the estrogen receptor Esr1 while animals assessed another mouse. This allowed activity to be aligned with the position and behavior of the social partner.

The selected recording follows an assessment-related cell when the stimulus animal is nearby. The animal view and aligned signal show the relationship between social investigation and neural activity. A population defined by a receptor can contain cells with different response patterns within the same encounter.

The study found representations of both the threatening animal and the context in which threat had occurred. Responses therefore depended on more than the immediate presence of another body. Prior experience could change which features of a social situation recruited the hypothalamic population.



Krzywkowski et al. (2020), Fig. 1.



Social assessment: nearby stimulus animal. Krzywkowski et al. (2020).

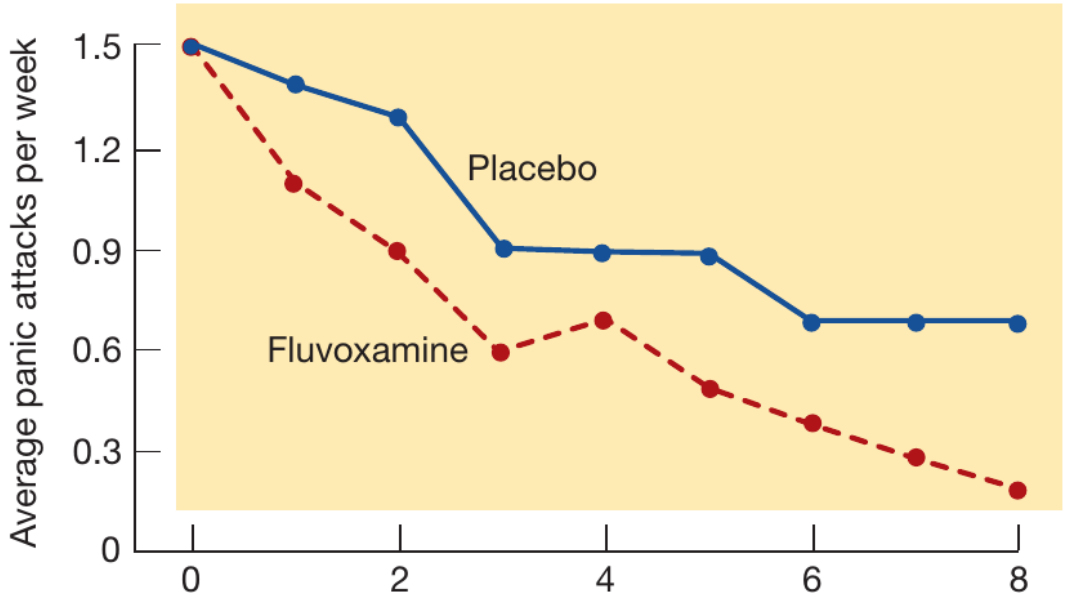
Play video

Panic attacks combine bodily and cognitive symptoms

A panic attack is an abrupt surge of intense fear or discomfort with symptoms such as a racing heart, breathlessness, dizziness, or trembling. The person may fear losing control or dying. The episode is different from a continuously elevated level of worry throughout the day.

Panic disorder involves recurrent unexpected attacks together with persistent concern or maladaptive changes in behavior related to them. Avoidance can develop around situations associated with bodily sensations or previous attacks. A single panic attack can occur in several conditions and does not by itself establish panic disorder.

Bodily signals and their interpretation can reinforce one another. A change in breathing may feel threatening, increasing arousal and further changing respiration. This feedback involves perception and physiology together, rather than requiring an external danger each time symptoms escalate.



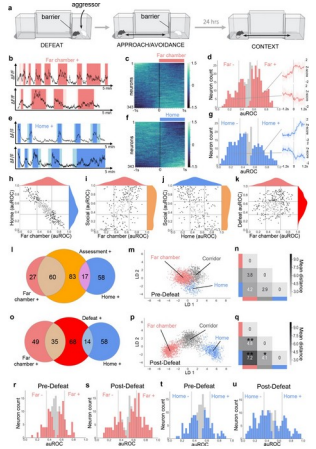
Carlson and Birkett (2022), Fig. 18.18, p.597.

Threat assessment includes spatial information

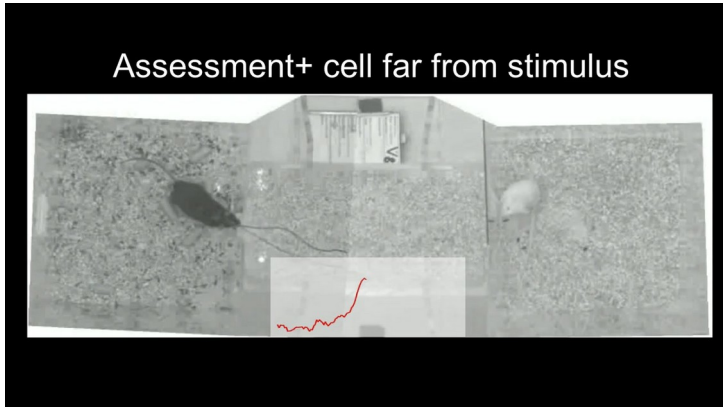
The social-threat study recorded cells with different spatial response patterns. The selected second example follows an assessment-related cell when the stimulus animal is farther away. It is a different cell and spatial relationship from the close-stimulus example, not another camera view of an identical neural event.

The activity trace is aligned with the encounter so that investigation and distance can be interpreted together. Neural responses depended on the social circumstances rather than general movement alone. The experiment therefore asked what information the cells represent while the mouse evaluates another animal.

After threatening experience, the population also responded to the associated setting. A context can continue to recruit defensive regulation when the original threatening individual is not immediately present. Such contextual responses provide a mechanism for experience to change later social approach and avoidance.



Krzywkowski et al. (2020), Fig. 2.



Social assessment: more distant stimulus animal. Krzywkowski et al. (2020).

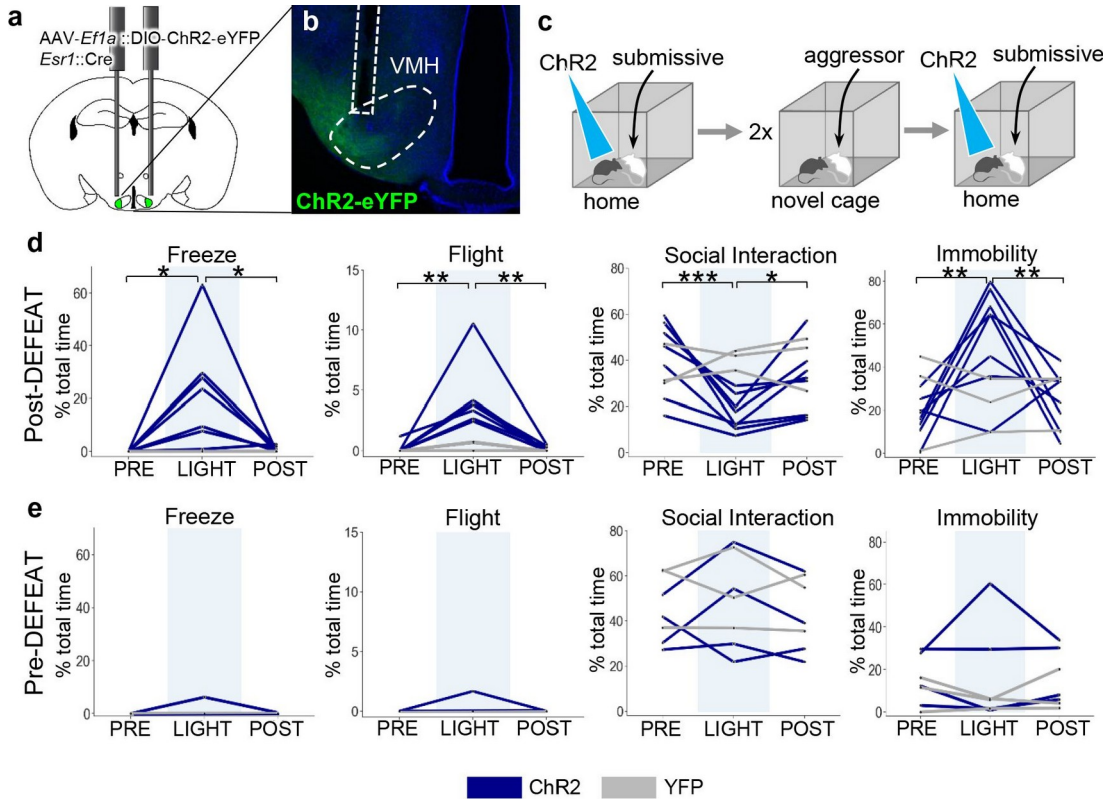
Play video

Agoraphobia centers on difficult escape or help

Agoraphobia involves marked fear or avoidance of situations in which escape may feel difficult or help unavailable if severe symptoms occur. Examples include public transport, crowded places, open areas, or being outside alone. The concern concerns the anticipated consequences of symptoms in that situation.

The person may endure the situation with intense distress or require a companion. Avoidance can progressively restrict travel and ordinary activities. This pattern can occur with panic disorder, but the two diagnoses describe different features and are not interchangeable names for one condition.

Learning can maintain the avoidance because leaving a feared situation rapidly reduces distress. That relief strengthens the tendency to avoid it again. Treatment can address the expectation of danger and the avoidance pattern through planned practice, rather than merely asking the person to suppress bodily sensations.



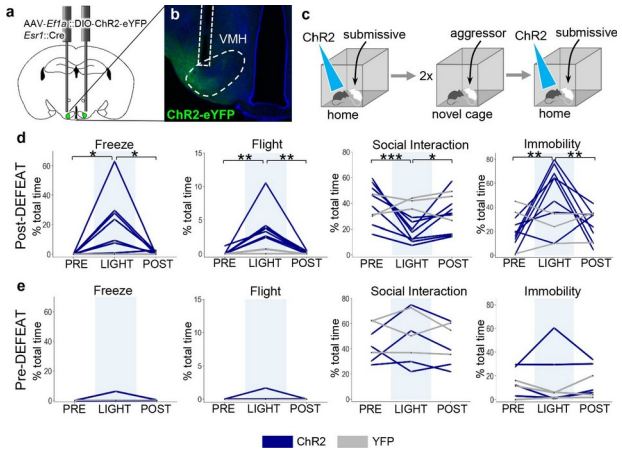
Krzywkowski et al. (2020), Fig. 5.

Experience changes the effect of hypothalamic stimulation

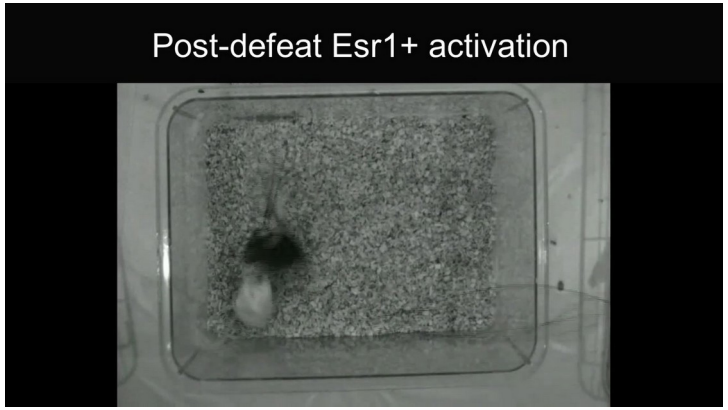
The social-threat researchers stimulated Esr1-expressing hypothalamic neurons before and after defeat experience. The same broad population did not produce an identical behavioral effect in both states. Experience changed how activation influenced the mouse's approach to the social situation.

After defeat, optical stimulation promoted defensive avoidance. The selected recording follows this post-defeat condition, with the stimulation period identified in the original source. Its interpretation depends on the preceding social experience rather than on illumination being inherently threatening.

This result assigns experience-dependent function to a defined hypothalamic population. Neural activation can produce a different outcome after the animal has learned about a threatening encounter. The experiment concerns mouse social defense, while human social anxiety additionally involves beliefs, evaluation, and impairment in social situations.



Krzywkowski et al. (2020), Fig. 5.



Post-defeat hypothalamic stimulation. Krzywkowski et al. (2020).

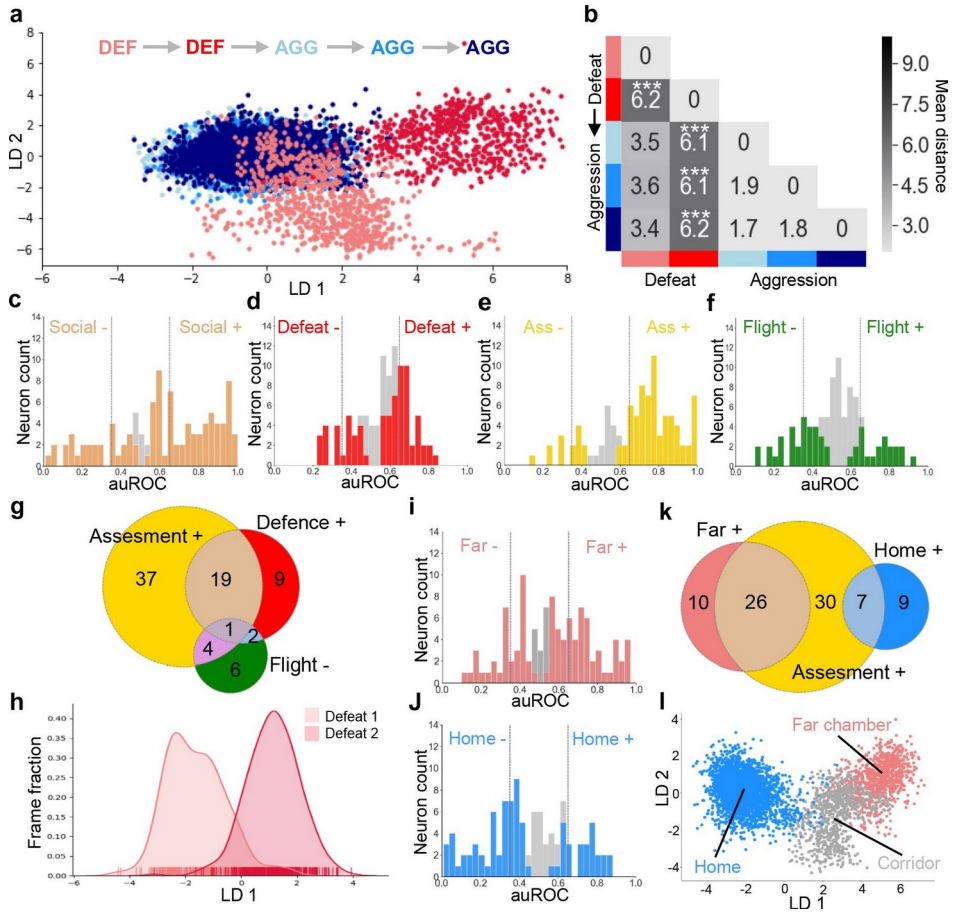
Play video

Worry and social evaluation define different patterns

Generalized anxiety disorder involves difficult-to-control worry across several areas of life. Restlessness, muscle tension, sleep disturbance, and problems concentrating can accompany it. The worry persists across circumstances rather than being limited to one specific object or a brief unexpected panic episode.

Social anxiety disorder centers on situations involving possible scrutiny or negative evaluation by others. Speaking, eating, or interacting in public can become distressing. The feared outcome concerns embarrassment or judgment, and avoidance can interfere with education, employment, and relationships.

These patterns can coexist with depression or other anxiety symptoms. Diagnosis identifies the dominant triggers and functional effects rather than assuming one brain region uniquely defines each disorder. Treatment targets the person's actual pattern of worry, threat expectation, and avoidance.



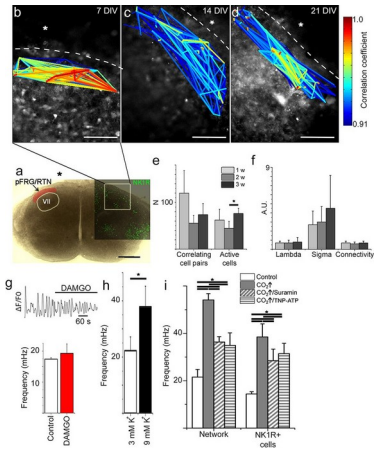
Krzywkowski et al. (2020), Fig. 4.

Carbon dioxide changes respiratory-network activity

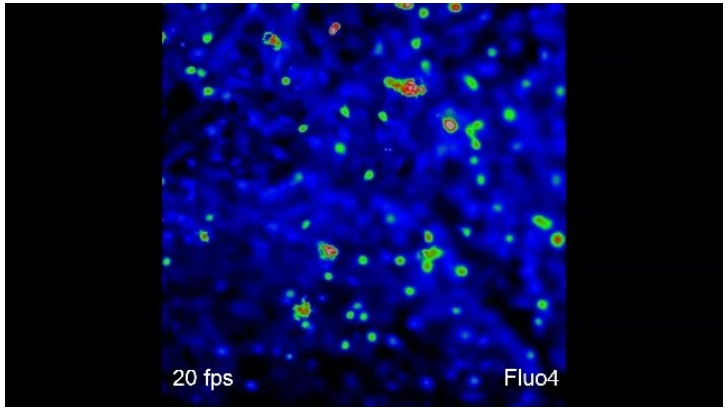
Carbon dioxide contributes to regulation of breathing because its accumulation changes chemical conditions, including acidity. Researchers maintained neonatal rodent brainstem tissue in culture to examine networks sensitive to these changes. The preparation preserved respiratory-related activity outside the intact animal.

The selected recording follows calcium fluorescence in the parafacial respiratory group and retrotrapezoid region. Increasing carbon dioxide raised activity in this chemosensitive network. The fluorescence follows cellular calcium changes rather than chest movement or a person's report of breathlessness.

Chemosensitive neurons and nearby astrocytes contribute to detecting the altered chemical environment. Their signals help change respiratory drive so that carbon dioxide can be removed. The cultured network identifies a physiological response to carbon dioxide, separate from the subjective experience of a human panic attack.



Forsberg et al. (2016), Fig. 9.



Carbon-dioxide response in brainstem culture. Forsberg et al. (2016).

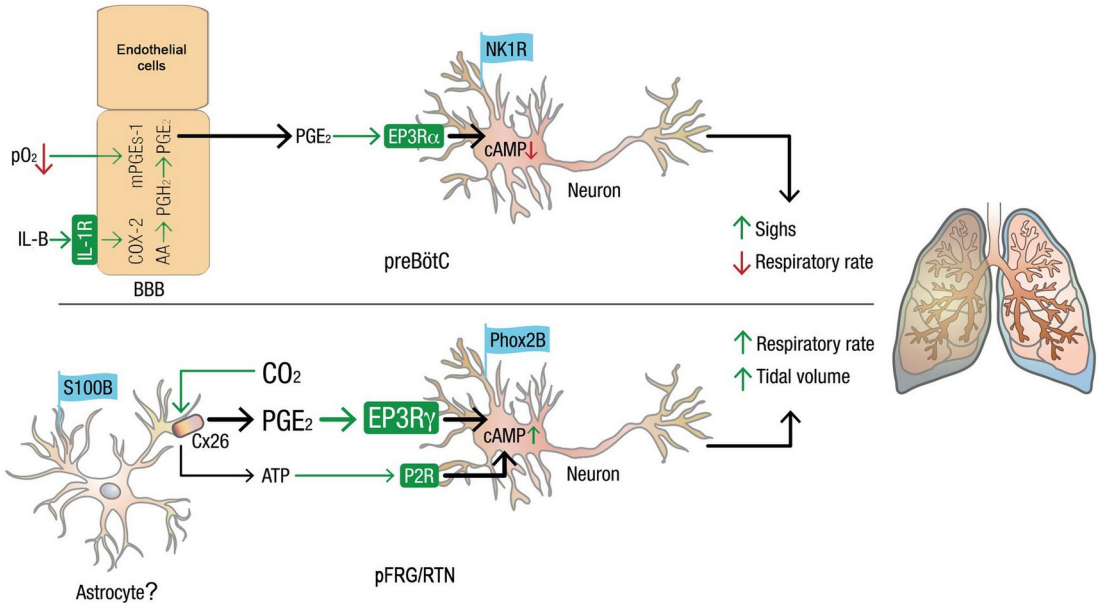
Play video

Internal sensations can become learned danger signals

Interoception is the sensing of conditions inside the body, including heartbeat, breathing, and visceral changes. These sensations provide useful information about physiological state. They can also become cues that predict danger when an individual learns to fear their occurrence.

Some people with panic symptoms respond strongly to respiratory challenges or sensations associated with rising carbon dioxide. The response involves both physiological detection and the meaning assigned to the sensation. Sensitivity to a laboratory challenge is not a complete explanation of every spontaneous attack.

Interoceptive exposure uses carefully planned experiences of feared sensations to support new learning. The aim is to revise catastrophic expectations and reduce avoidance under appropriate clinical guidance. This learning process differs from directly suppressing every normal change in heart rate or breathing.



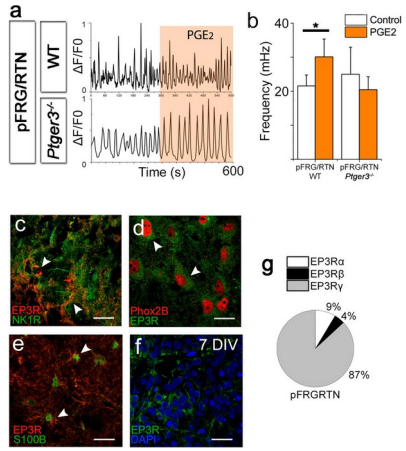
Forsberg et al. (2016), Fig. 14.

PGE2 affects respiratory regions differently

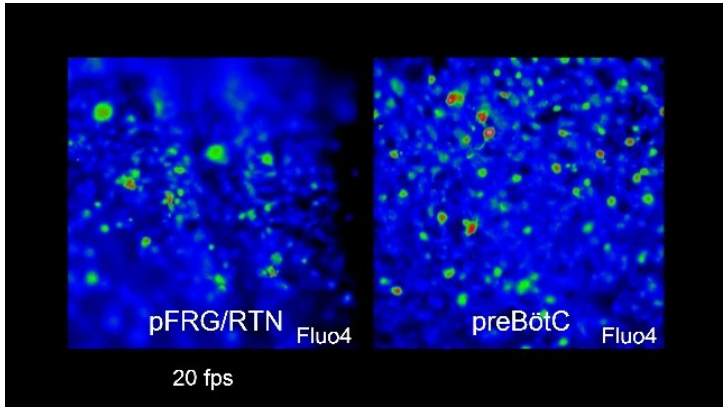
Prostaglandin E2 is a chemical messenger produced during several physiological and inflammatory processes. The brainstem-culture study applied it while imaging respiratory-related networks. Comparing regions allowed the investigators to test whether one messenger changed all respiratory activity in the same direction.

The original paired recording shows the parafacial region on the left and the pre-Bötzinger complex on the right. After prostaglandin E2 is added, activity increases in the parafacial network while the pre-Bötzinger network is inhibited. The two regions therefore show different responses in the same experimental setting.

The study also examined the EP3 receptor, through which prostaglandin signaling contributes to respiratory regulation. These cellular effects concern neonatal respiratory circuitry and inflammatory modulation. They are not evidence that prostaglandin administration is a treatment for panic disorder.



Forsberg et al. (2016), Fig. 12.



PGE2 effects in two respiratory networks. Forsberg et al. (2016).

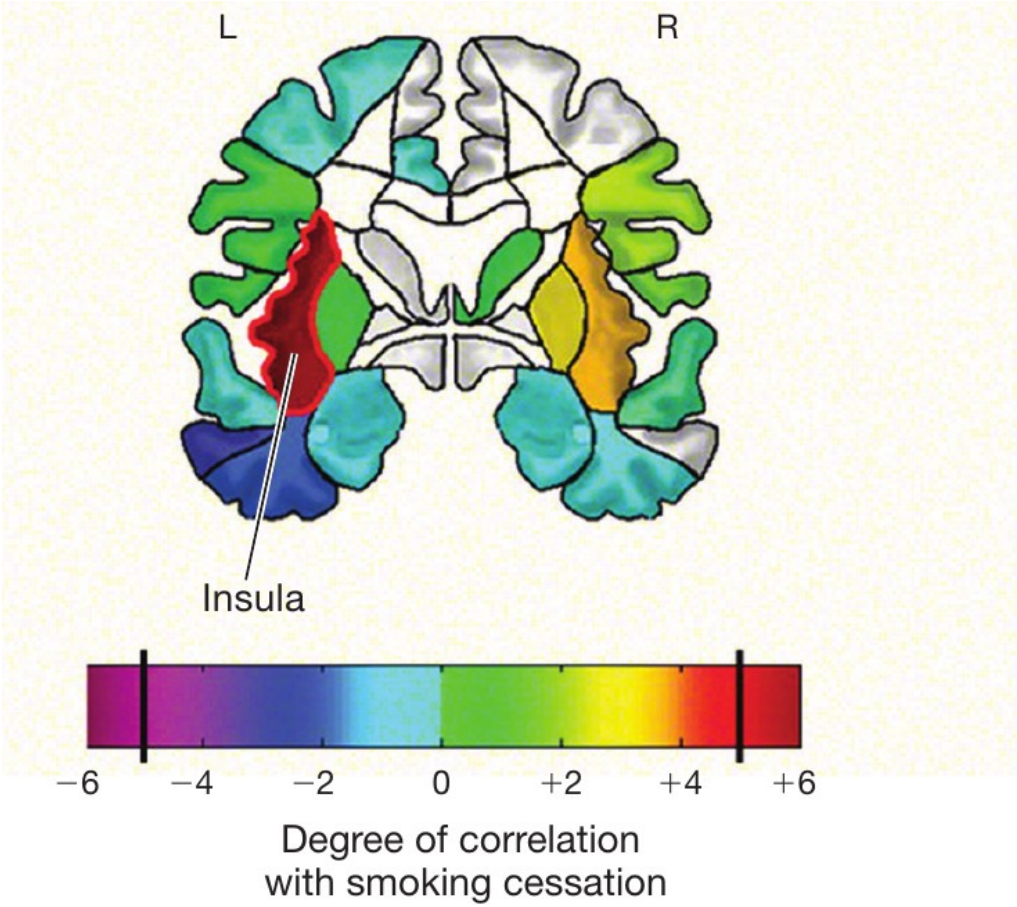
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The insula represents bodily state during emotion

The insular cortex receives information about bodily conditions and contributes to awareness of internal sensations. Activity in this system can accompany heartbeat perception, breathing discomfort, and emotional experience. A bodily signal therefore becomes part of a representation that includes its significance to the person.

The amygdala participates in evaluating biologically important events and influencing defensive outputs. Its interactions with cortical and brainstem systems can change attention and autonomic responses. Panic-related bodily arousal can recruit this distributed network without one structure independently generating every symptom.

Predicting what a sensation means can alter the response to it. A faster heartbeat during exercise and a similar sensation interpreted as imminent danger can lead to different behavior. The sensory event, learned expectation, and resulting action must therefore be described separately.



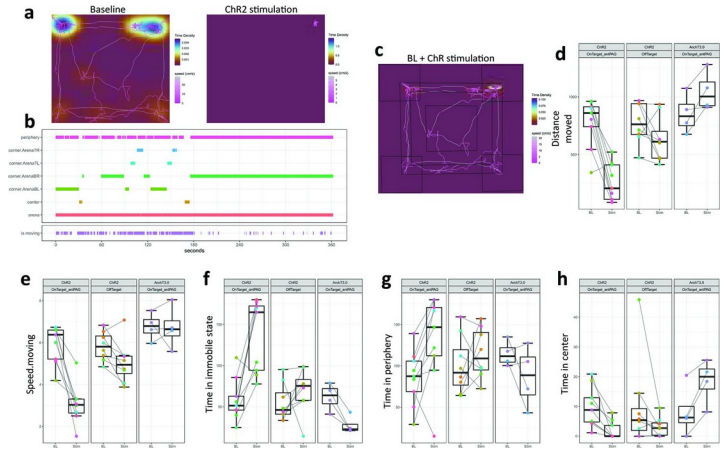
Carlson and Birkett (2022), Fig. 19.18, p.624.

A hypothalamic projection can produce immobility

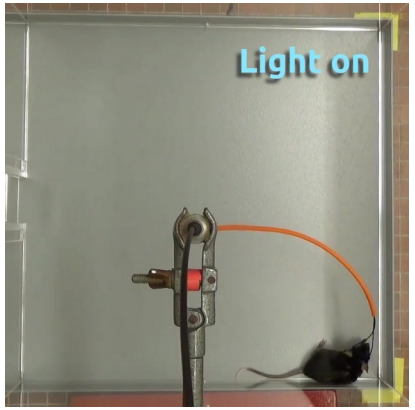
Researchers examined hypothalamic neurons identified by expression of Foxb1, a gene-regulating protein used to distinguish the population. Some of these neurons project to the dorsolateral periaqueductal gray, a midbrain region involved in defensive responses. Optical stimulation targeted their axon terminals in that region.

In the selected mouse example, stimulation produced marked immobility while the head retained some movement. The original light-on interval identifies when the terminals were activated. The action is an experimentally evoked defensive pattern rather than evidence that the mouse experienced a human panic episode.

The study separately measured breathing and heart-related responses. Activating whole-cell populations and activating one projection did not produce identical physiological patterns. The location at which the circuit is manipulated therefore determines which outputs are recruited during the experiment.



Cola et al. (2024), Fig. 3.



Foxb1-terminal stimulation with residual head movement. Cola et al. (2024).

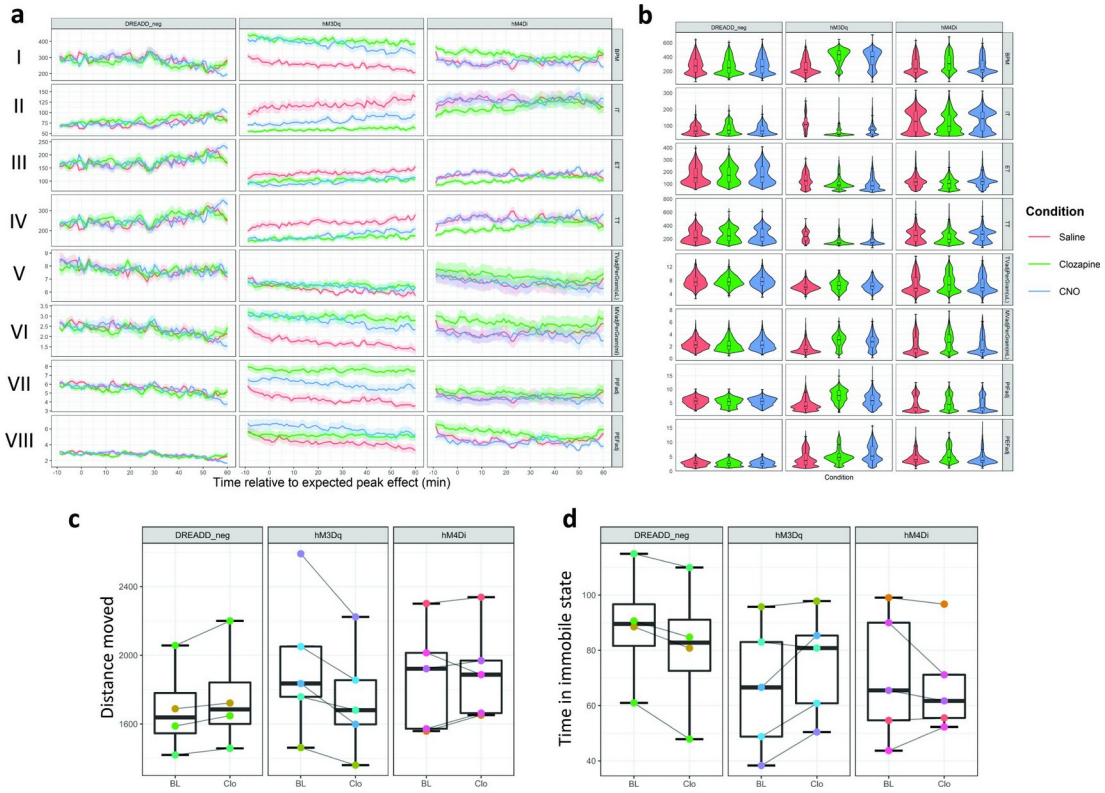
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Cell bodies and axon branches recruit different outputs

Chemogenetic activation uses an engineered receptor to change the activity of selected cells after a matching compound is given. In the Foxb1 study, activating hypothalamic cell bodies increased breathing rate. This manipulation could recruit the cells' several outputs rather than only one terminal field.

Optical activation of terminals in the rostral dorsolateral periaqueductal gray instead produced immobility accompanied by slowing of the heart. The more selective manipulation identified a projection capable of organizing that response. It did not require every output of the parent neurons to be stimulated simultaneously.

A neuron can send branches to several targets, and each target participates in different local circuitry. Activating the cell body therefore need not reproduce the effect of activating only one branch. The experiment separates a population's overall influence from a specific descending pathway.



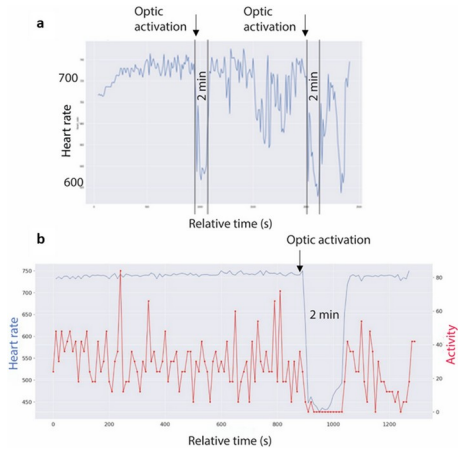
Cola et al. (2024), Fig. 2.

The degree of immobility varies across examples

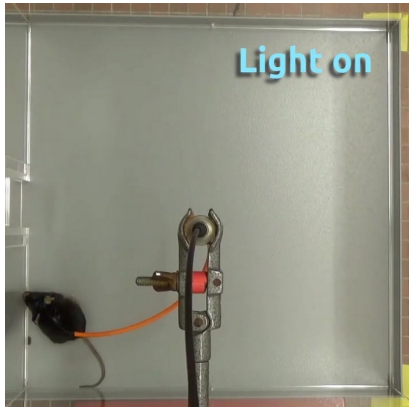
A second original Foxb1-projection recording shows a mouse becoming more completely immobile during terminal stimulation. It demonstrates the same defined projection under a different representative response. The two examples do not represent separate clinical treatments or two different diagnoses.

The original source preserves the light-on period and playback timing. Comparing movement before, during, and after stimulation identifies when the behavioral change occurs. Remaining still must be distinguished from loss of the ability to move, because the experiment measures a circuit-driven defensive state.

Physiological recordings in the paper also identified bradycardia, a decrease in heart rate, during this projection-specific response. Defensive behavior can therefore include cardiovascular slowing rather than always producing acceleration. The combination depends on the circuit and form of defense being recruited.



Cola et al. (2024), Fig. 6.



Foxb1-terminal stimulation with greater immobility. Cola et al. (2024).

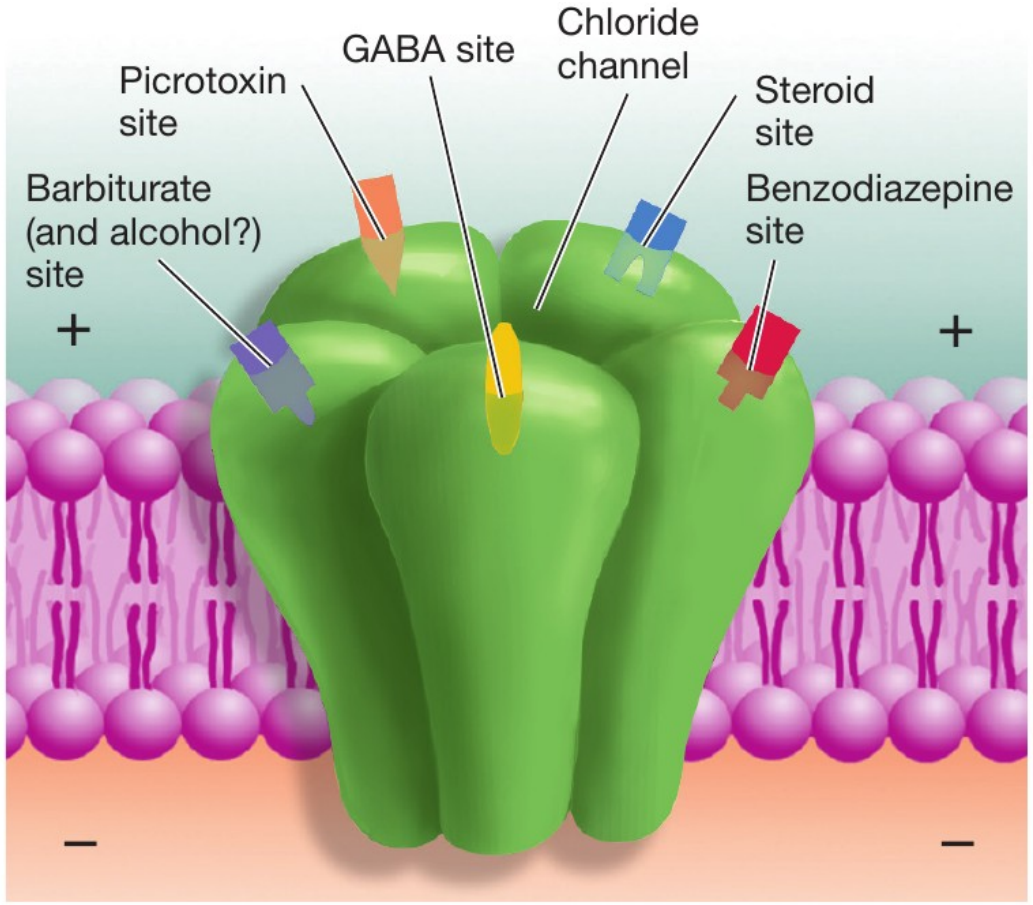
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Benzodiazepines strengthen existing GABA inhibition

GABA is a major inhibitory neurotransmitter in the brain. At GABA-A receptors, its binding opens an ion channel that usually reduces the likelihood of neuronal firing in mature circuits. Benzodiazepines bind another site on certain GABA-A receptors and enhance the receptor's response to GABA.

These medicines can reduce anxiety rapidly, but their effects also include sedation and impaired coordination. With repeated use, tolerance and physical dependence can develop. The mechanism does not erase a learned threat association or directly teach that a feared situation is safe.

Combining sedating drugs can dangerously suppress central nervous system functions. Clinical decisions therefore consider duration of use, other medicines, and the person's circumstances. A receptor-level account explains how inhibition is increased, while prescribing requires assessment of benefits and risks for the individual.



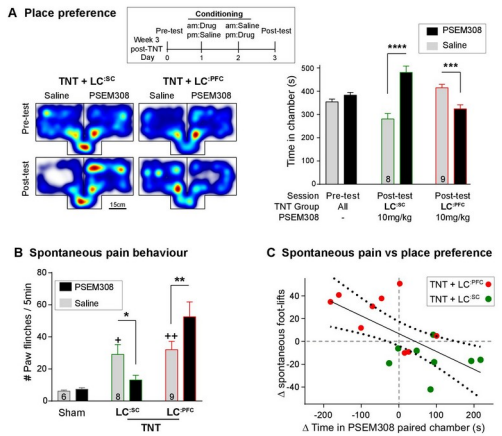
Carlson and Birkett (2022), Fig. 18.17, p.597.

A spinal norepinephrine pathway can reduce pain behavior

A separate locus-coeruleus study distinguished neurons by the targets of their axons. Researchers activated the population projecting toward the spinal cord in a neuropathic-pain model. Neuropathic pain arises from injury or dysfunction in the sensory nervous system rather than a new external injury at each response.

The selected source recording compares behavior after activation of this descending population. Spontaneous pain-related behavior was reduced in the tested condition. The manipulated neurons belong to a spinally projecting module, not the separate population projecting to prefrontal cortex in the same paper.

This result differs from the acute stress-related spinal astrocyte mechanism discussed earlier. The preparation, recruited cells, and pain state are different. Norepinephrine can influence several local targets, so its effect cannot be assigned one universal direction independently of the pathway and experimental condition.



Hirschberg et al. (2017), Fig. 7.



Spinally projecting LC activation in neuropathic pain. Hirschberg et al. (2017).

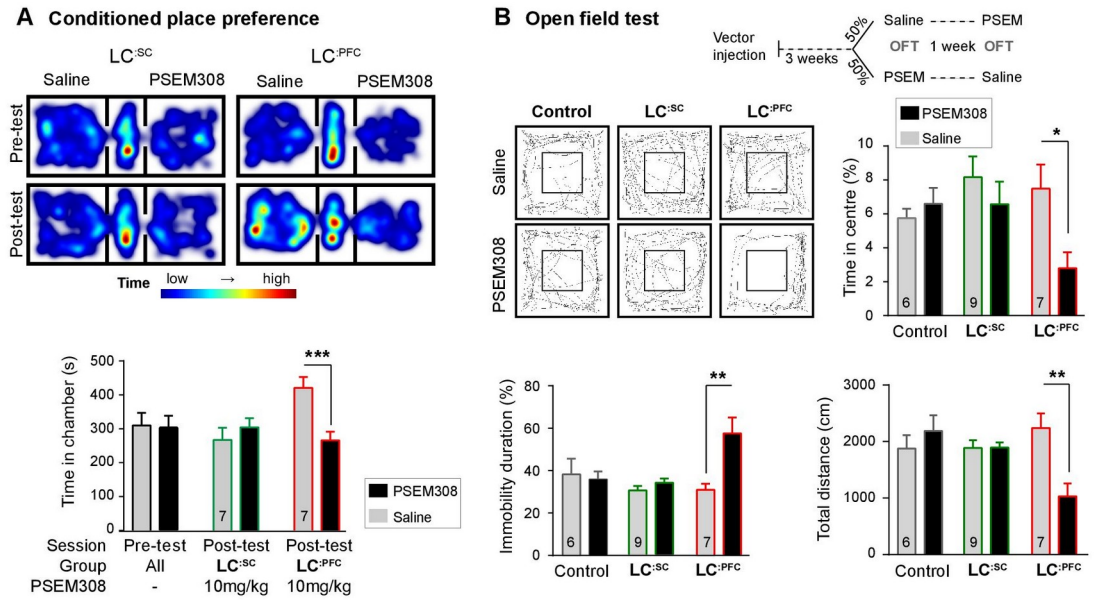
Play video

Locus-coeruleus modules have different behavioral effects

The locus coeruleus sends norepinephrine-containing axons to many regions, but its cells are not interchangeable. The modularity study distinguished populations projecting to spinal cord and prefrontal cortex. This anatomical separation allowed their contributions to pain and aversive behavior to be tested individually.

Activating the spinally projecting population reduced pain-related responses in the tested model. Activating the prefrontal-projecting population produced a different pattern involving aversion and anxiety-related behavior. One brainstem transmitter system therefore contained outputs with different functional consequences.

The distinction depends on where norepinephrine is delivered and which local receptors and cells receive it. A drug that changes the whole system can recruit more than one module. Projection-specific experiments identify those components before a broad pharmacological effect is interpreted as a single process.



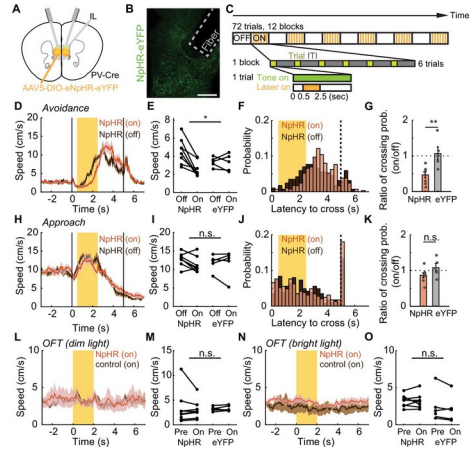
Hirschberg et al. (2017), Fig. 4.

Local inhibitory neurons can help initiate avoidance

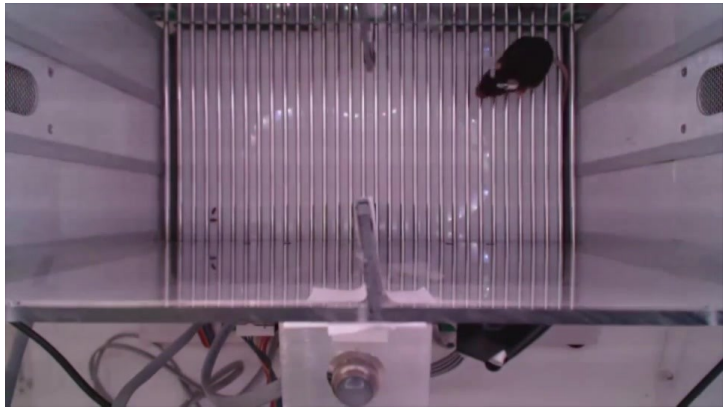
Infralimbic cortex is a medial prefrontal region in mice that contributes to flexible responses to threat. Researchers examined its parvalbumin-expressing interneurons, inhibitory cells that act within local circuits. They recorded their activity while mice learned to move toward safety during a warning cue.

The interneurons became active during threat-related movement, including after an early aversive experience before avoidance was fully learned. Their activity did not increase in the same way during ordinary movement or movement toward a reward. The response therefore concerned the need to overcome defensive freezing.

Optically suppressing these cells prolonged freezing and delayed avoidance, as the original comparison demonstrates. Inhibition within cortex can consequently support an active response through its effects on local circuit organization. It is not accurate to assume that increasing inhibitory-cell activity must suppress every useful behavior.



Ho et al. (2025), Fig. 5.



Infralimbic PV suppression delays avoidance. Ho et al. (2025).

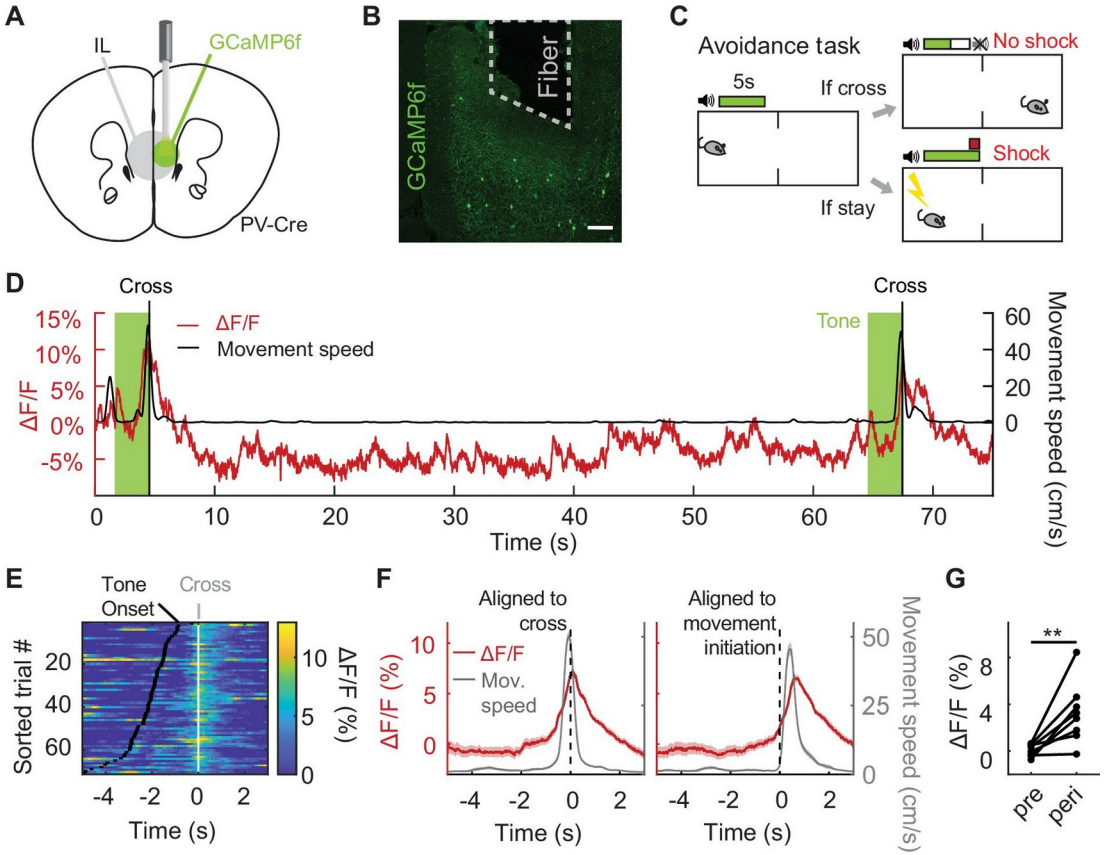
Play video

Learning and medicines act on different timescales

Cognitive behavioral therapies address interpretations, habits of attention, and actions that maintain anxiety. Exposure provides opportunities to revise threat predictions, while other strategies help the person respond differently to worry or symptoms. Improvement involves learning and practice across relevant situations.

SSRIs reduce serotonin reuptake soon after reaching their target, but clinical improvement usually develops through later changes over time. Receptor regulation and circuit adaptation occur after the initial transporter effect. Immediate molecular binding and sustained symptom reduction are therefore different stages of treatment.

Researchers have also tested medicines intended to assist extinction-related learning, including the NMDA-receptor partial agonist D-cycloserine. Its role concerns plasticity during learning rather than simple sedation. Such an approach depends on what is learned during the treatment session, not merely whether the compound was taken.



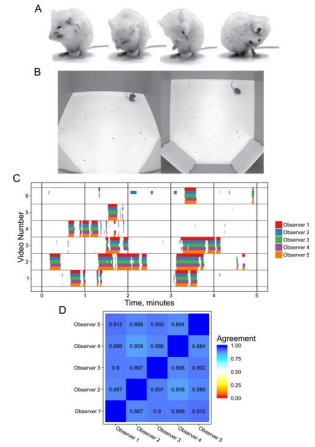
Ho et al. (2025), Fig. 1.

Compulsions differ from ordinary repeated actions

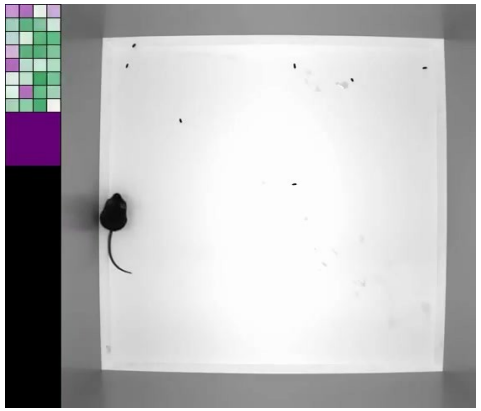
Obsessions are intrusive, unwanted thoughts, urges, or images that cause distress. Compulsions are repetitive actions or mental acts performed in response to an obsession or a rigid rule. In obsessive-compulsive disorder, or OCD, these experiences consume time or substantially interfere with functioning.

A compulsion can briefly reduce distress, strengthening the tendency to repeat it when the concern returns. The person may recognize that the action is excessive while still finding it difficult to resist. The reinforcing effect concerns relief from distress rather than necessarily obtaining pleasure.

Grooming is also a normal animal behavior, as the black-mouse recording demonstrates. It becomes an experimental measure of excessive repetition only when its amount, organization, and consequences are assessed. Ordinary grooming alone does not establish a mouse equivalent of intrusive human thoughts.



Geuther et al. (2021), Fig. 1.



Ordinary grooming: black mouse. Geuther et al. (2021).

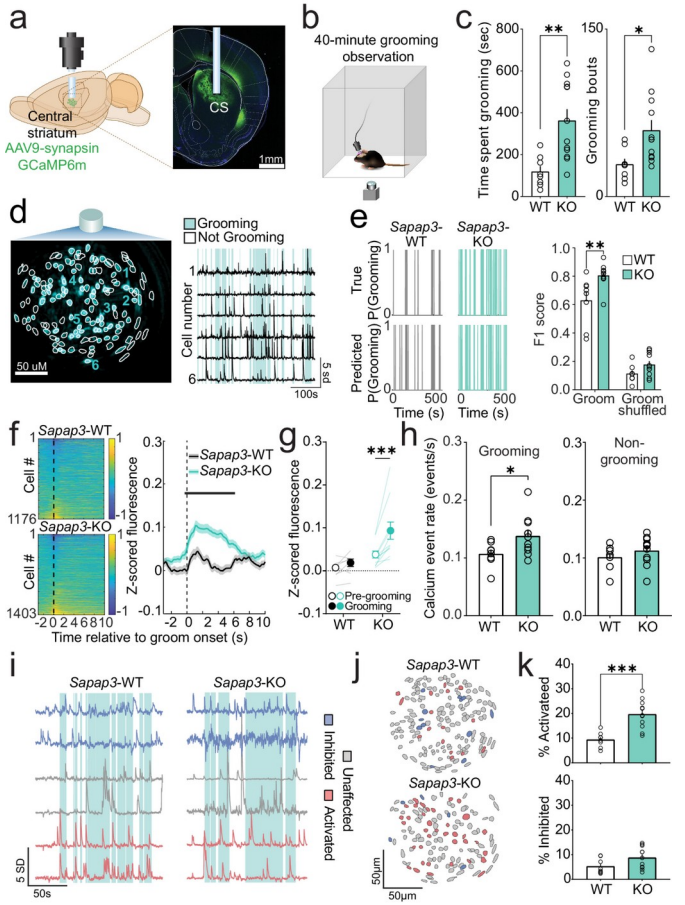
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Synaptic proteins can alter corticostriatal function

The striatum receives input from the cerebral cortex and participates in selecting and organizing actions. Its projection neurons integrate these inputs before influencing other basal-ganglia structures. Changes at these synapses can alter the circumstances in which an action is initiated or repeated.

SAPAP3 is a scaffolding protein located at excitatory synapses. A scaffold helps hold signaling components in useful positions near the receiving membrane. Removing the corresponding gene in mice disrupts synaptic organization and produces excessive grooming in a widely studied experimental model.

A gene deletion affects a biological preparation rather than reproducing every feature of a human diagnosis. The model permits direct tests of particular cells and pathways during repeated behavior. Human OCD risk involves many genetic and environmental influences rather than absence of this one protein in every patient.



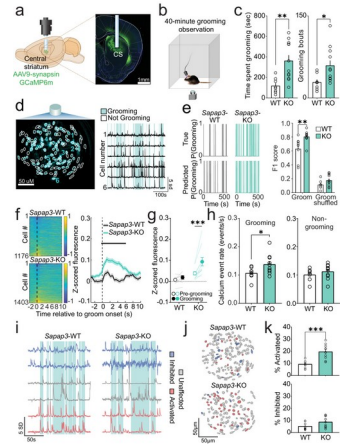
Piantadosi et al. (2024), Fig. 1.

Striatal activity rises around excessive grooming

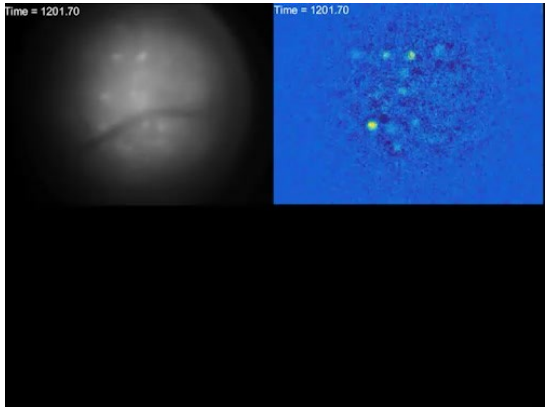
Researchers imaged calcium activity in the central striatum of mice lacking Sapap3 and their ordinary littermates. A miniature microscope followed labeled neurons while grooming was scored. The deficient mice initiated more grooming bouts and spent more total time grooming, without a corresponding increase in typical bout duration.

Activity in the overall striatal population increased around grooming onset. The population signal could help identify when grooming occurred, and this information was stronger in the deficient mice. The association concerned the timing of an observed action rather than a direct measurement of intrusive thoughts.

The original movie displays calcium imaging beside the extracted cellular signals. The extraction separates candidate cell regions from the raw image so their fluorescence can be followed. It is a baseline imaging demonstration, not a recording of a drug reversing the behavior.



Piantadosi et al. (2024), Fig. 1.



Central-striatal baseline calcium imaging. Piantadosi et al. (2024).

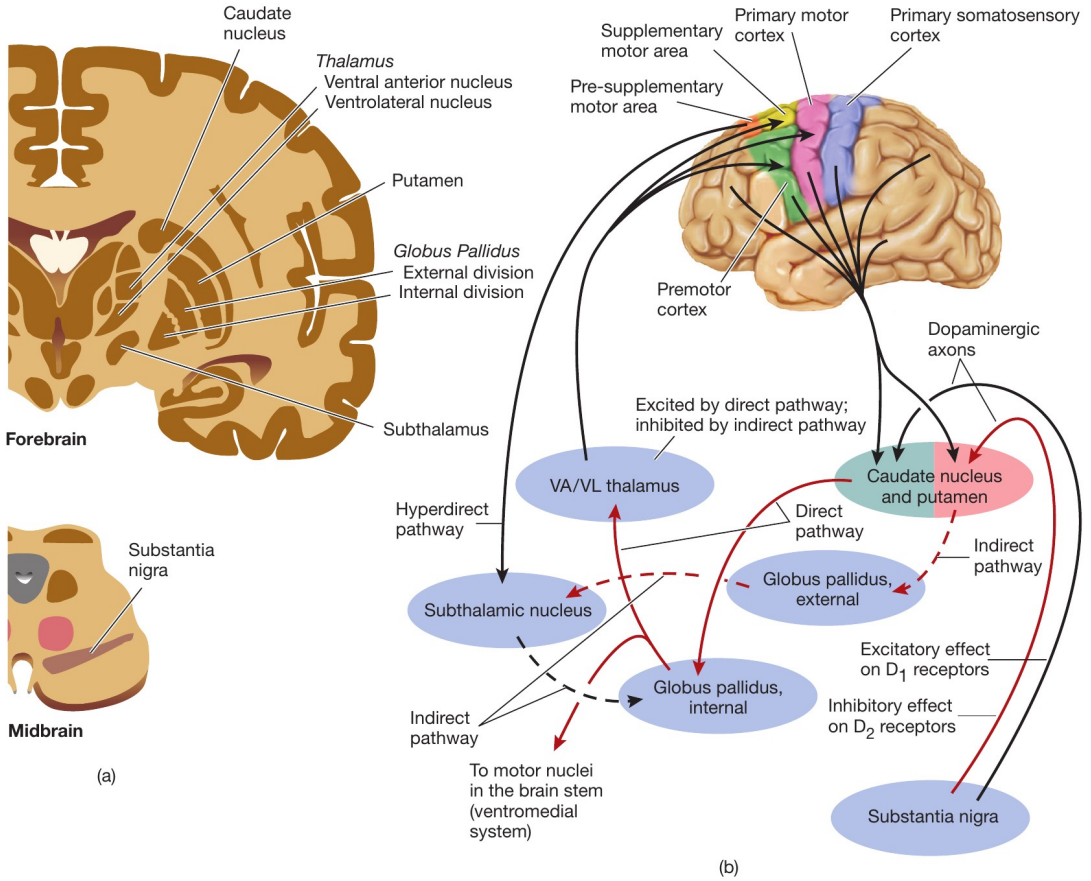
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Basal-ganglia pathways regulate actions through inhibition

Striatal projection neurons release GABA and inhibit their targets. Direct-pathway neurons project toward basal-ganglia output structures, while indirect-pathway neurons influence them through additional stages. Their effects depend on which inhibitory connection is changed at each step of the circuit.

Disinhibition occurs when an inhibitory cell is itself inhibited, reducing the restraint on its target. A pathway can therefore facilitate a downstream process even though its own neurons release an inhibitory transmitter. Calling the direct pathway simply excitatory would obscure this sequence of effects.

D1 and D2 dopamine receptors help distinguish major striatal populations, although biological categories are not perfectly exclusive. Both populations can be active during movement. Explaining an excessive action requires measuring their actual activity rather than assuming that all repetition comes from one pathway becoming stronger.



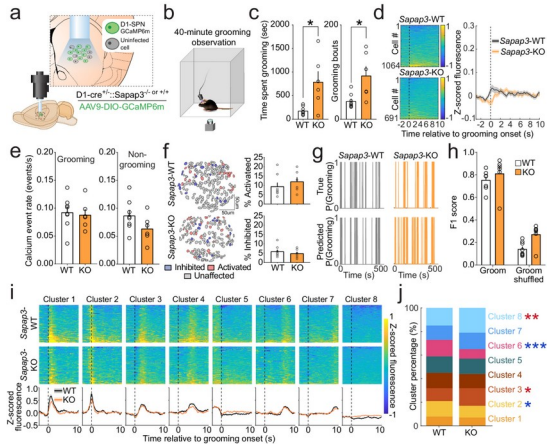
Carlson and Birkett (2022), Fig. 18.22, p.602.

Direct-pathway cells did not explain the hyperactivity

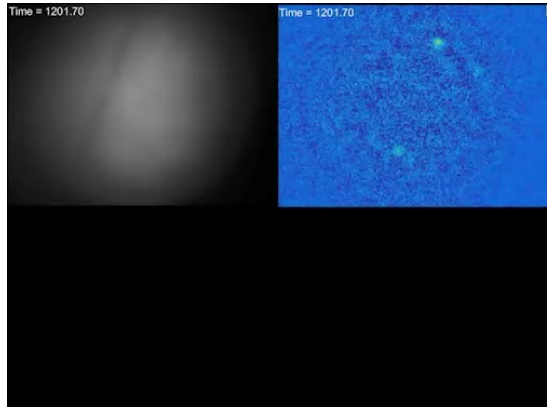
The Sapap3 study next restricted calcium imaging to D1-expressing striatal neurons. This population largely corresponds to the direct pathway. The researchers tested the prediction that its activity would account for the larger overall signal observed when deficient mice began excessive grooming.

D1-cell activity did not show the expected elevation at grooming onset in deficient mice. Event rates during grooming also did not explain the overall striatal hyperactivity. A plausible circuit prediction therefore failed when the investigators measured the proposed cell population directly.

The original D1-cell movie demonstrates the targeted imaging preparation and extracted signals. It does not compare treated and untreated states or itself establish a group difference. The paper's aligned population analyses provide the finding that direct-pathway hyperactivity was not the explanation in this model.



Piantadosi et al. (2024), Fig. 2.



D1-cell baseline calcium imaging. Piantadosi et al. (2024).

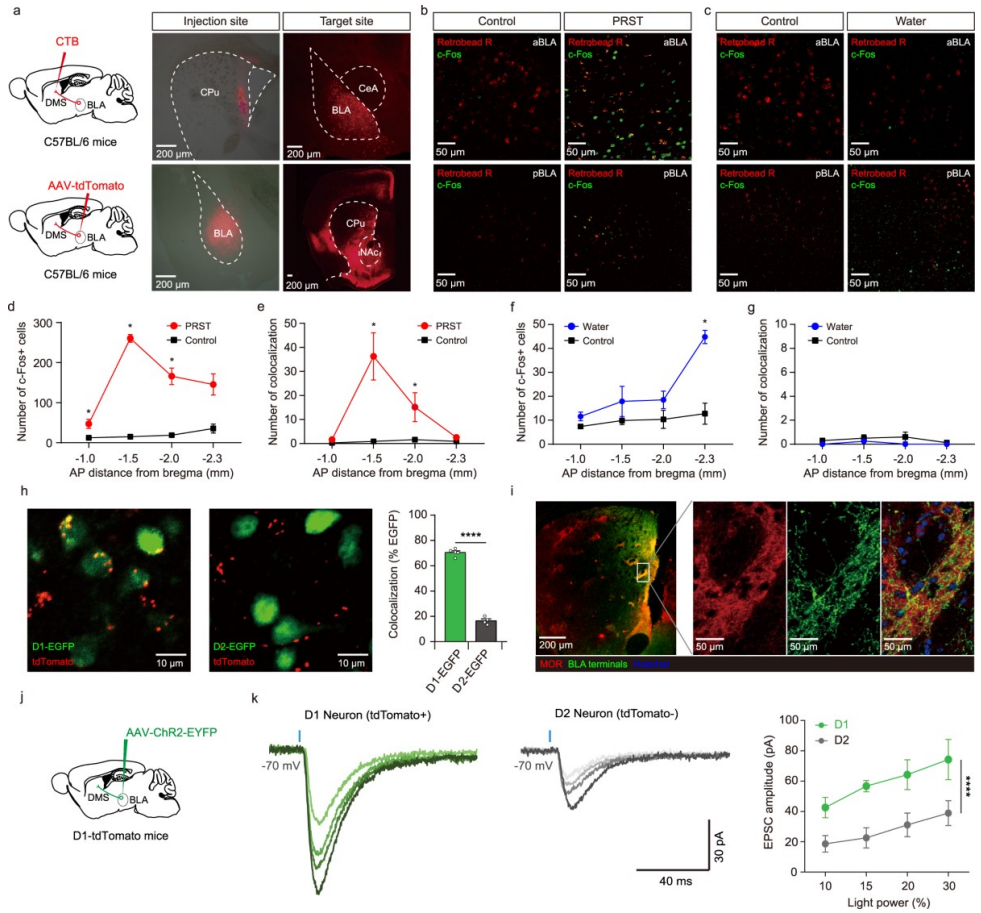
Play video

Frontal-striatal loops contribute to persistent actions

The orbitofrontal cortex helps represent outcomes and changes in their significance. It communicates with striatal regions that participate in action selection, and basal-ganglia output returns information through the thalamus. These recurring pathways are commonly called frontal-striatal loops.

A loop can sustain or repeatedly recruit an action when its component signals fail to update appropriately. In OCD, imaging and experimental work implicate networks involving frontal cortex, striatum, and related regions. This does not mean that all patients have the same anatomical lesion or one uniformly overactive cell type.

Outcome evaluation and action repetition are also separable processes. A person can judge a ritual unnecessary while continuing to feel driven to perform it. Circuit explanations must account for the persistence of the action as well as the representation of its expected consequences.



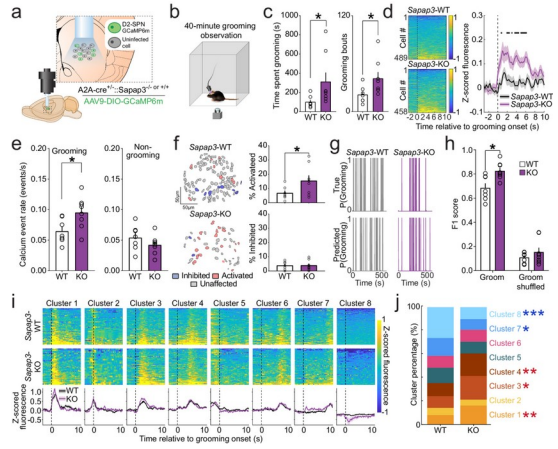
Lee et al. (2024), Fig. 2.

Indirect-pathway cells were overactive during grooming

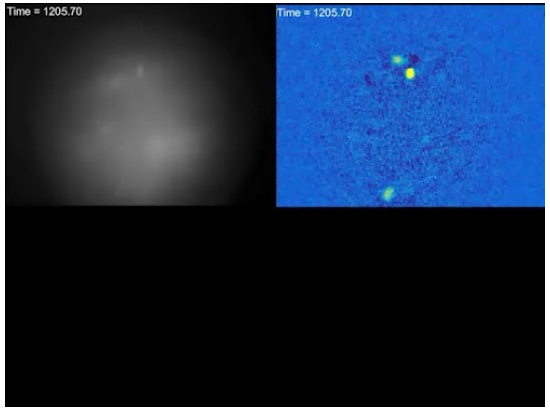
The investigators used an A2a genetic marker to image striatal neurons enriched for D2 receptors and indirect-pathway projections. This separated them from the D1 population measured previously. Calcium activity was aligned with the onset of grooming in ordinary and Sapap3-deficient mice.

The deficient mice showed elevated activity in the D2-enriched population when grooming began. Calcium-event rates were also higher during grooming, while the same increase was not present during non-grooming periods. The abnormality therefore depended on both cell type and behavioral state.

The original A2a-targeted movie shows the baseline imaging signals used for this population. It is not an optogenetic inhibition session. These measurements identified an unexpectedly active population that could subsequently be tested with a separate causal manipulation.



Piantadosi et al. (2024), Fig. 3.



A2a-targeted baseline calcium imaging. Piantadosi et al. (2024).

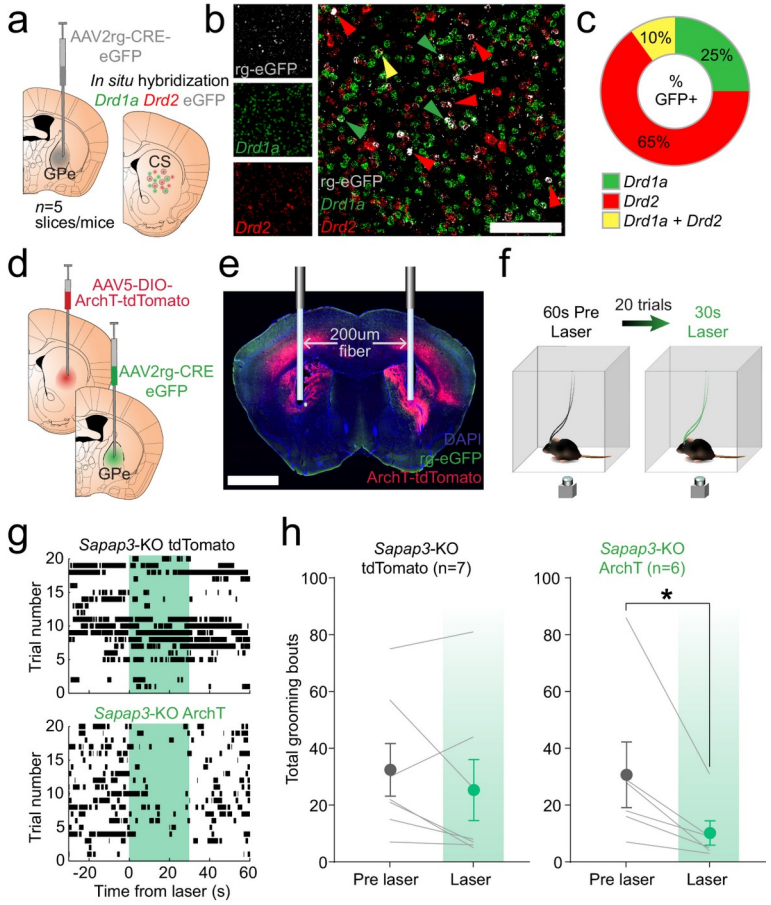
Play video

Inhibiting striatopallidal cells reduces excessive grooming

To test causality, researchers targeted striatal neurons that project to the external globus pallidus. A backward-traveling viral tool identified the cells by their axon destination. This strategy enriched for the indirect pathway while defining the manipulated population anatomically.

The cells expressed ArchT, a light-sensitive protein that reduces activity during illumination. In Sapap3-deficient mice, inhibiting this projection-defined population reduced excessive grooming. Control animals expressing only the comparison marker did not show the same reduction during the light periods.

The manipulation interrupted activity in a population identified by the preceding imaging experiments. It therefore went beyond observing that activity and behavior occur together. The result concerns grooming in this genetic mouse model, with the tested circuit providing a specific target rather than a universal explanation of human compulsions.



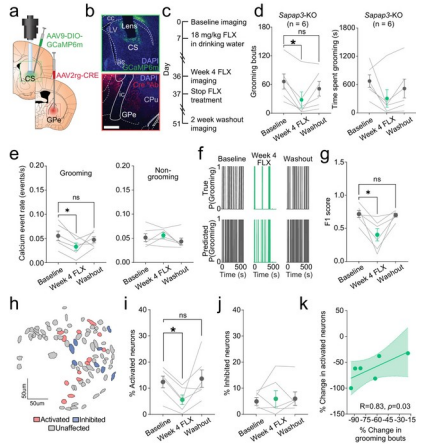
Piantadosi et al. (2024), Fig. 4.

Fluoxetine changes grooming-related circuit activity

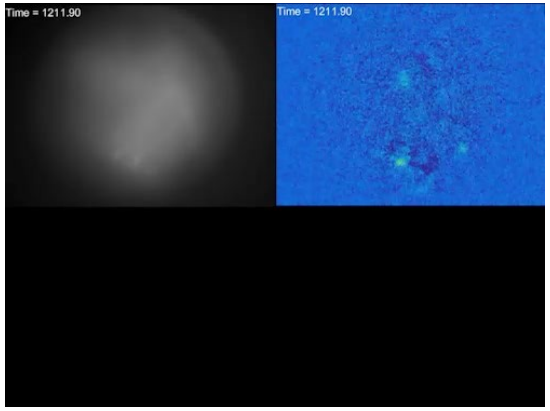
The same study followed striatopallidal activity during fluoxetine treatment. Fluoxetine blocks serotonin reuptake, but its later effects can include adaptation in circuits outside the immediate transporter site. The researchers therefore measured both grooming and the activity associated with it over time.

A longer treatment period reduced the number of grooming bouts and reduced abnormal grooming-related activity. Total grooming time did not show the same treatment effect. The distinction matters because initiating fewer bouts and shortening every bout would represent different behavioral changes.

The selected movie is a baseline recording from projection-defined striatopallidal cells in an ordinary mouse. It demonstrates the imaged population, not the fluoxetine response itself. The treatment findings come from longitudinal behavioral and calcium analyses presented in the original figure.



Piantadosi et al. (2024), Fig. 5.



Projection-defined baseline calcium imaging. Piantadosi et al. (2024).

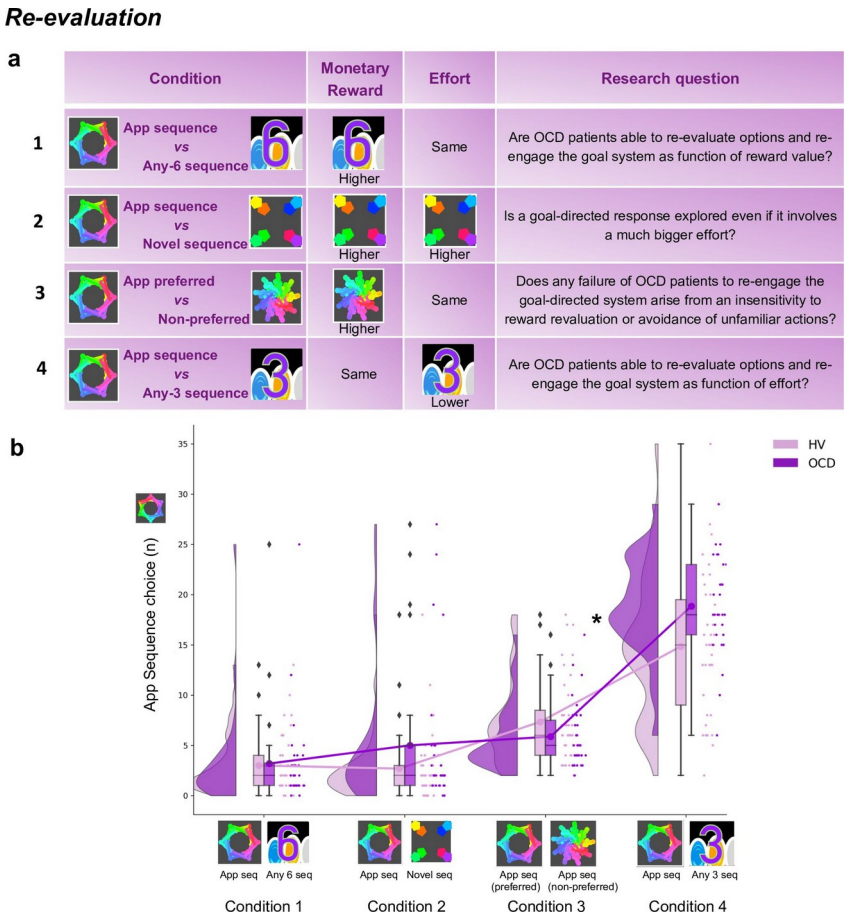
Play video

Response prevention interrupts relief-driven repetition

Negative reinforcement strengthens an action because the action removes something unpleasant. In a compulsion, performing a ritual can briefly reduce anxiety or the sense that something is wrong. That relief can strengthen repetition even when the person knows the ritual does not provide realistic protection.

Exposure and response prevention, or ERP, combines contact with a feared cue and practice refraining from the ritual. The person can learn that distress changes without completing the compulsion and that feared outcomes do not follow as expected. Treatment is planned with appropriate clinical support.

Avoidance and reassurance can also preserve the ritual's apparent necessity by preventing new learning. ERP targets this maintaining cycle rather than simply demanding willpower. The clinical outcome concerns distress, time spent in rituals, and restored functioning, not merely suppression of all repeated movement.



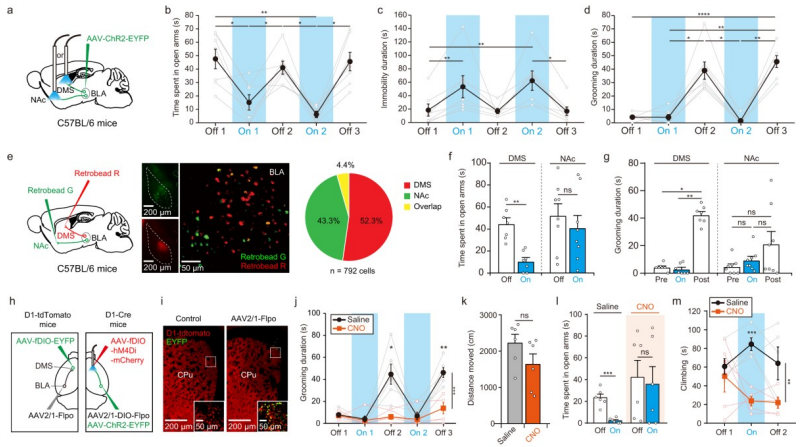
Banca et al. (2024), Fig. 8.

Amygdala input to striatum can evoke defensive behavior

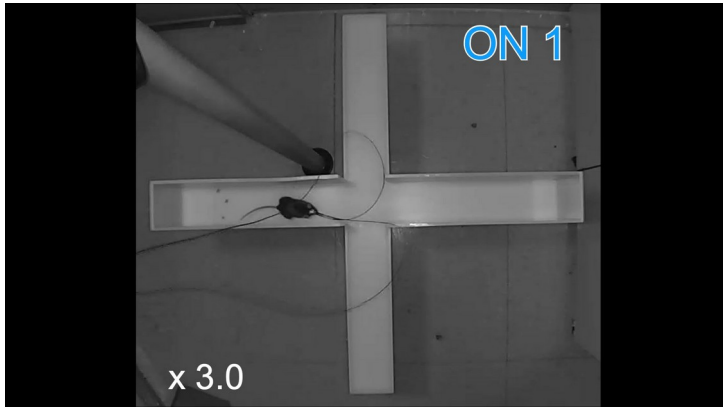
A separate mouse study examined projections from the basolateral amygdala to dorsomedial striatum. Many of the projecting cells also responded to predator-related stress. Their terminals mainly contacted D1-expressing striatal neurons in the measured pathway.

Optical stimulation of this projection produced anxiety-related avoidance followed by excessive grooming. The selected elevated-plus-maze recording follows behavior during stimulation. The maze has exposed and enclosed arms, allowing the distribution of exploration to be measured during the circuit manipulation.

The movie therefore demonstrates behavior in an anxiety-related task, not a person's experience of an obsession. The study combined such task measures with direct manipulation of a defined amygdala-striatal pathway. The behavioral sequence identified how recruiting that input could alter both avoidance and later repetition.



Lee et al. (2024), Fig. 3.



BLA-DMS stimulation in the elevated plus maze. Lee et al. (2024).

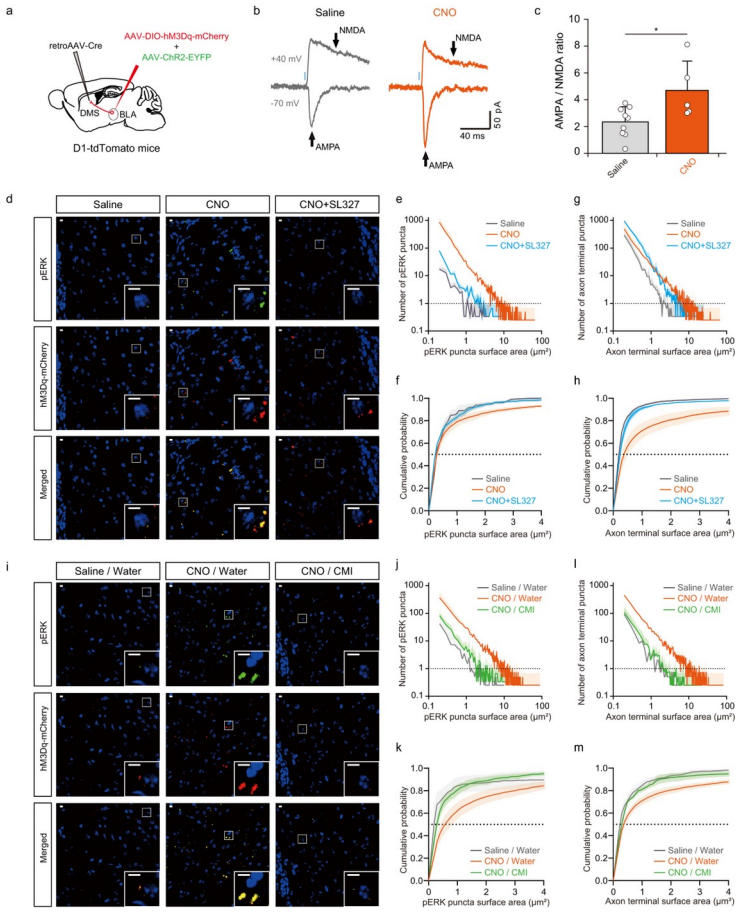
Play video

Repeated activation produces lasting synaptic changes

The amygdala-striatal researchers repeatedly activated the projection across days. The mice later displayed a wider set of repetitive and compulsive-like actions. The persistence beyond an individual light pulse prompted the investigators to examine whether the pathway's synapses had changed.

Measurements showed lasting enhancement of transmission at amygdala inputs to dorsomedial striatum. The study also examined molecular changes related to ERK signaling, a pathway in which enzymes add phosphate groups to regulate downstream proteins. These changes accompanied the enduring behavioral phenotype.

Interventions directed at the altered pathway reduced compulsive-like behavior in the tested model. The evidence therefore included both persistent synaptic change and manipulation of its contribution. A temporary increase in activity and a lasting change in synaptic strength describe different stages of the experimental process.



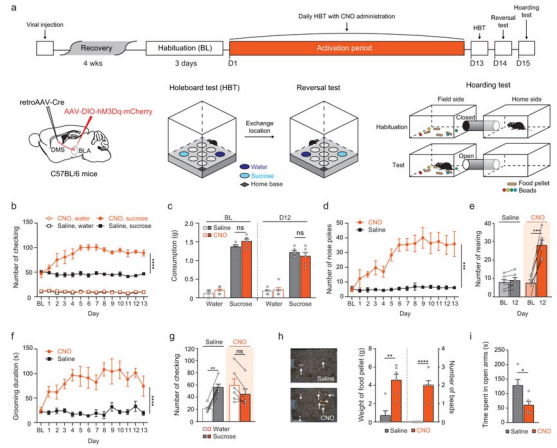
Lee et al. (2024), Fig. 6.

A holeboard task separates checking from exploration

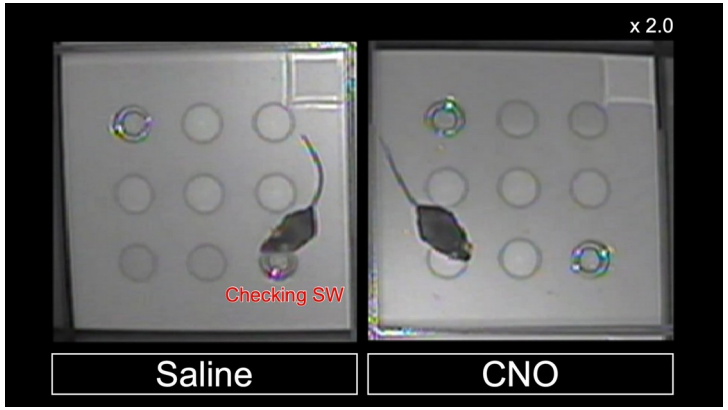
The amygdala-striatal study used a holeboard containing several openings and objects. Investigators scored returns to locations, nose pokes, and grooming as distinct actions. A repeated visit was therefore measured within the actual arrangement rather than inferred from the animal remaining in one area.

The original comparison shows behavior under saline and circuit-activation conditions using an engineered receptor. The activation condition increased compulsive-like checking and related actions in the study. The movie's labels identify these task events as the mouse explores the board.

Nose poking, checking, and grooming are not interchangeable measurements. They engage different movements and can change independently. Scoring each action allowed the researchers to describe a broader phenotype after repeated pathway activation, rather than claiming that one increase in grooming explained every observed behavior.



Lee et al. (2024), Fig. 4.



Holeboard behavior during circuit activation. Lee et al. (2024).

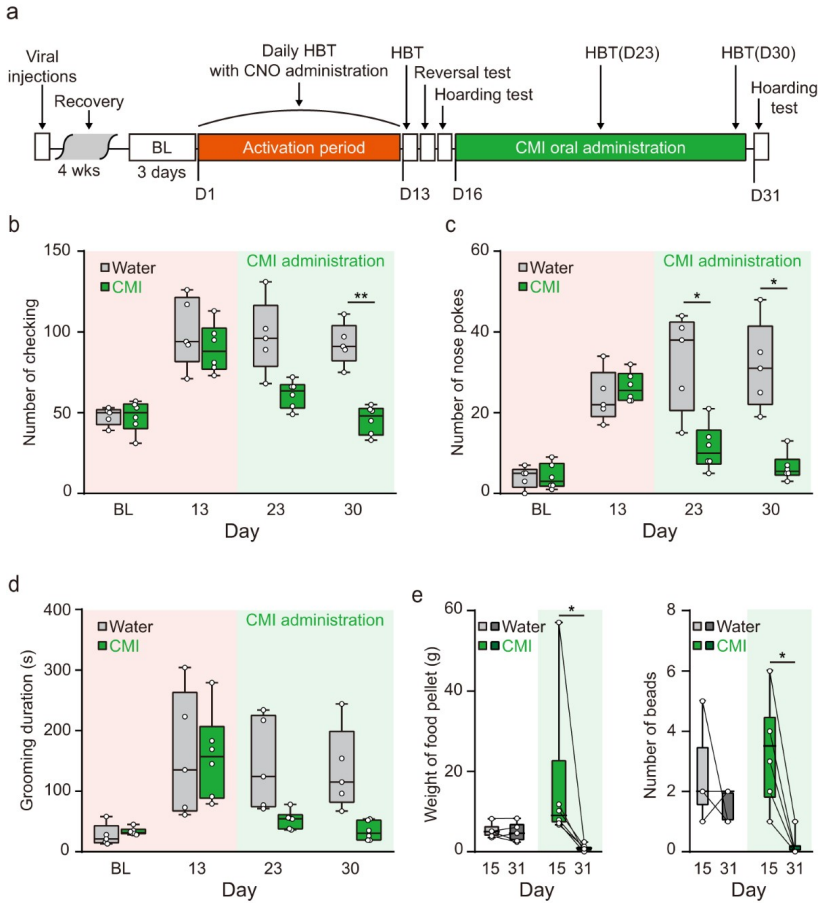
Play video

Serotonin-reuptake inhibitors can reduce OCD symptoms

SSRIs block the transporter that normally returns released serotonin to the presynaptic cell. More serotonin consequently remains available outside cells after release. The medicine changes transmitter clearance; it does not directly force every serotonin receptor into its active state.

Clomipramine is another medicine with strong serotonin-reuptake inhibition used in OCD treatment. Its additional receptor actions contribute to a different adverse-effect profile. A shared effect on serotonin clearance does not make all medicines identical in their full biological effects.

Clinical improvement often develops gradually and may require sustained treatment. This delay separates the immediate transporter action from later adaptation in neural systems controlling symptoms. ERP and medication can address different components of the maintaining process and may be combined according to clinical circumstances.



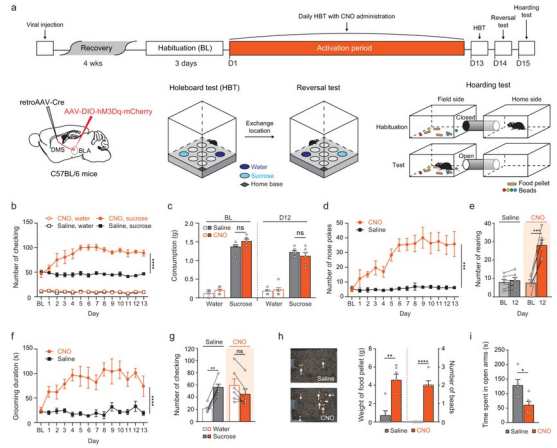
Lee et al. (2024), Fig. 5.

Repeated activation changes hoarding-related actions

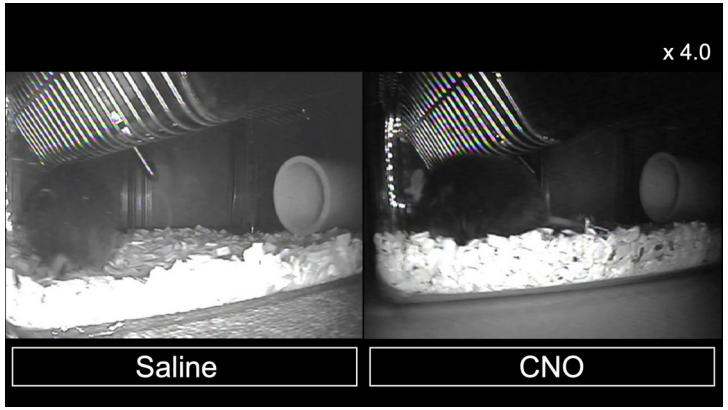
The amygdala-striatal study also examined behavior in a hoarding-related task. The original recording compares saline with activation of the engineered receptor in the defined circuit. It preserves the source's marked playback speeds and the visible periods of grooming and task-directed movement.

The activated condition produced abnormal repetitive behavior within this task as part of the broader phenotype. The outcome was not established solely by where the mouse stood. Investigators scored the relevant actions and compared them with the control condition after the repeated circuit manipulation.

Hoarding-related behavior in mice is an operational task measure, distinct from the beliefs and impairment used to assess hoarding disorder in people. The experiment tests whether one pathway manipulation changes several forms of repeated action. Each task supplies a separate behavioral observation within that circuit model.



Lee et al. (2024), Fig. 4.



Hoarding-task behavior during circuit activation. Lee et al. (2024).

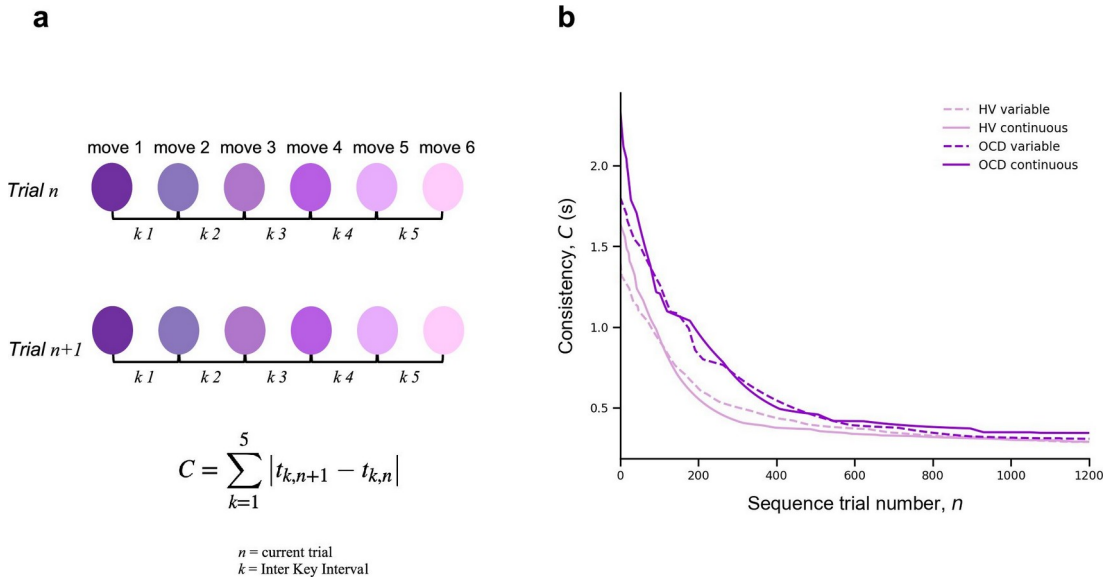
Play video

OCD does not require faster objective habit formation

A human study trained people with OCD and comparison participants on action sequences using a smartphone app. Repetition allowed individual finger movements to become an efficient sequence, or action chunk. The researchers tracked learning and used an objective criterion to assess increasing automaticity.

Both groups developed procedural skill and automatic performance at similar rates. Participants with OCD nevertheless reported stronger subjective habitual tendencies. Feeling that an action is habitual therefore did not require a measurable acceleration of habit formation in this particular task.

The investigators subsequently changed the consequences associated with actions to test later choices. Both groups remained sensitive to monetary revaluation. That result distinguished acquiring an efficient repeated sequence from continuing to choose it when the relevant outcome or effort requirement changed.



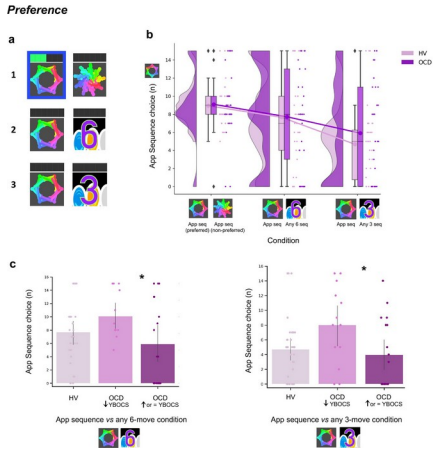
Banca et al. (2024), Fig. 4.

Familiar sequences can acquire greater intrinsic value

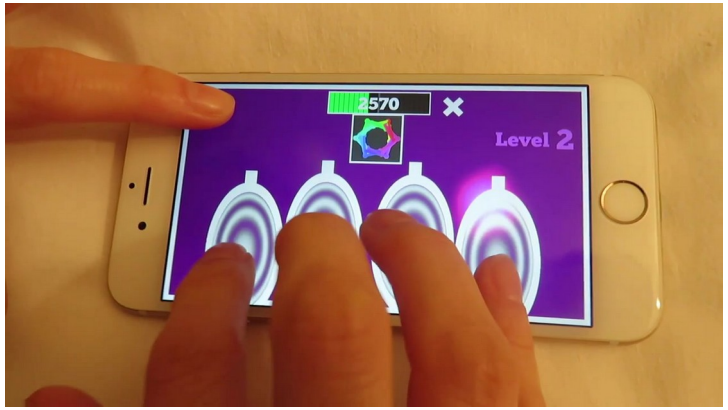
After app training, participants chose between established sequences and new goal-directed actions. When choices depended on physical effort, participants with OCD showed a stronger preference for familiar trained sequences. This differed from the preserved response to changes in monetary feedback.

The original app demonstration shows the sequence task rather than identifiable patient behavior. Repeated practice organizes separate taps into a fluent action pattern. The study then tested how much participants valued that familiar pattern when choosing between actions with different demands.

The authors proposed that familiar actions can acquire greater intrinsic value in OCD. Intrinsic value means the action itself becomes attractive beyond its external reward. This proposed mechanism concerns preference for established routines, rather than assuming that every compulsion results from universally faster habit learning.



Banca et al. (2024), Fig. 7.



Motor Sequencing App demonstration. Banca et al. (2024).

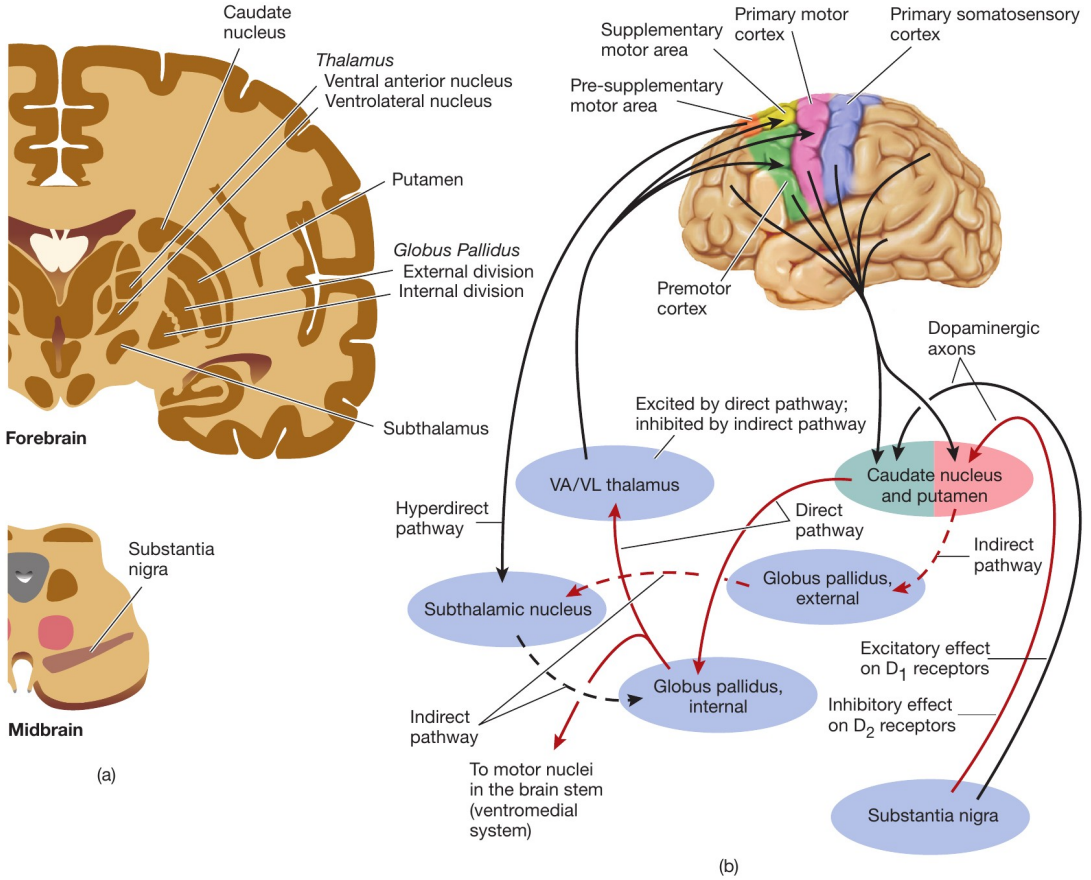
Play video

Brain stimulation targets severe persistent symptoms

Deep brain stimulation delivers electrical pulses through implanted electrodes to selected brain targets. In severe treatment-resistant OCD, stimulation can influence activity in networks involving frontal and basal-ganglia regions. The effect depends on target location and how the connected circuits respond.

Ablative procedures instead create a localized lesion intended to interrupt pathological network activity. A lesion and adjustable stimulation are different interventions even when they concern nearby pathways. Both require specialist assessment because altering these networks can also affect functions beyond the intended symptom.

Noninvasive stimulation methods act through fields applied outside the skull and are also used or investigated in selected settings. Their targets and supporting evidence differ from implanted stimulation. Circuit experiments help define candidate mechanisms, while clinical studies establish symptom outcomes and functional benefit in people.



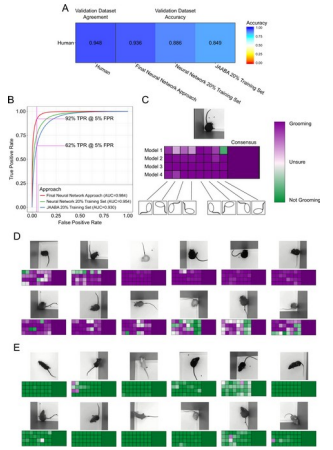
Carlson and Birkett (2022), Fig. 18.22, p.602.

Automated scoring distinguishes grooming bouts

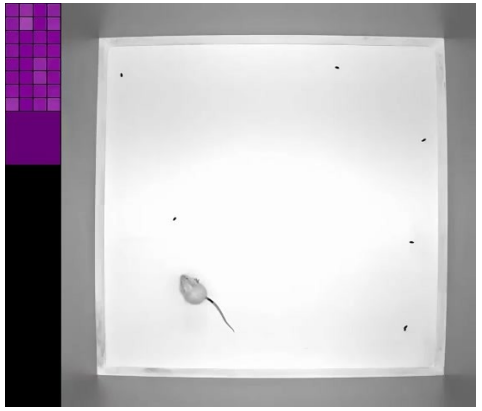
Researchers developed a computer-vision method for recognizing mouse grooming from ordinary behavior videos. The classifier learned visual features associated with the action and was compared with human scoring. Automated recognition made it possible to follow grooming consistently across many recordings.

The selected albino-mouse example demonstrates classification in a different visual background from the earlier black-mouse example. Coat color changes the image presented to the algorithm, not the definition of OCD. The study tested recognition across genetically diverse mice rather than treating appearance as a psychiatric phenotype.

Accurate scoring separates bout number, duration, and total grooming time. Those measurements can reveal different effects of genes or interventions, as the striatal experiments demonstrated. The automated method supplies the behavioral measurement; the biological experiment must still establish why a particular pattern changes.



Geuther et al. (2021), Fig. 3.



Grooming classification: albino mouse. Geuther et al. (2021).

Play video

Key takeaways

Stress responses have distinct routes. Sympathetic nerves, adrenal catecholamines, and the HPA axis operate through different structures and time courses; negative feedback helps limit glucocorticoid exposure.

Context changes defensive action. The same learned danger can produce freezing, retreat, flight, or social avoidance; score the actual action and identify the setting before assigning a mechanism.

Vulnerability and exposure are separate stages. Development, pre-existing anatomy, and later experience can each influence persistent symptoms; an anatomical difference measured afterward does not identify when it began.

Extinction is new learning. Safe experience changes the expression of a learned prediction without necessarily deleting the original association; context and prior training affect later retrieval.

Compulsions involve specific maintaining processes. Relief can reinforce rituals, familiar actions can acquire value, and cell-specific striatal changes can promote repetition; ordinary grooming is not itself OCD.

Treatments act at different biological levels. ERP changes learned expectations and ritual responding, reuptake inhibitors alter serotonin clearance, and stimulation changes network activity; evaluate clinical symptoms and functioning separately from animal circuit outcomes.